

Informational Dissipative Structures

A Relative-Information Framework for Thermodynamically Consistent Emergent Structure Across Scales

Laurent Caraffa

Univ. Gustave Eiffel, IGN-ENSG, LaSTIG

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Abstract

We propose Informational Dissipative Structures (IDS), a framework for modelling physical systems from finite observations. A declared observation channel decides which candidate laws are distinguishable and which share one signature; the primitive is relative information to an explicitly declared reference, read on three distinct spaces. On process paths, the time-reversal reference yields irreversibility, and another declared path reference yields the corresponding change of path law. On observable laws, every channel obeys an exact visible/residual ledger, whose second-order differential is the Fisher-information ledger of the quotient of experimentally identifiable laws. On probability measures over that quotient, Bayesian updating is characterized by a least-relative-information variational principle, and the Wasserstein slope of relative entropy defines a calibrated geometry of belief revision, the relative-entropy slope geometry (RESG). The kernel is free of any graph or clique structure: finite resolution is an ordered family of channels, each level carrying a resolution ledger, visible plus residual. Along an exhaustive tower the ledger recovers the fine values of relative information, Fisher information and Bayesian information monotonically, and so does the calibrated radius under the radial hypotheses; an almost-sufficient projection is almost isometric; a finite description closes when its predictive ledger says so. Irreversibility obeys the same bookkeeping: under a quotient that commutes with time reversal a level reads a share of the path irreversibility, the entropy production of a thermal realization under local detailed balance, and an exhaustive refinement recovers it; a realization and its coarse grainings are thermodynamically consistent across scales when these refinement laws are joined to closed declared port balances at every level. Graphs, cells and symbols enter only in realizations of finite resolution. The Brownian gyrator, a driven ring and a metastable double well separate path irreversibility, local relative-information balance, thermodynamic entropy production and predictive closure. The contribution is the assembly of classical ingredients into one reference-explicit reduction calculus with a common information ledger.

1 Introduction

The modelling problem. We model a phenomenon from observations that are partial and noisy: to its observer, a phenomenon is first of all uncertain. A cell seen through a microscope, a reactor seen through its sensors, a machine seen through its logs, a climate seen through its stations: what a model receives is a finite reading, and what it seeks is the law behind the reading. Every model is therefore a reduction, and the first question of this paper is what a reduction costs.

Three families of models. Three families answer that reading. A mechanistic model writes equations for the state and integrates them, and its native criterion is the distance between its trajectories and the data. A graphical model factorizes a joint law over the cliques of a graph, so that interaction becomes locality and inference becomes the fusion of local potentials (Lauritzen, 1996), and its native criterion is a likelihood. A learned model fits a map from data: a world model compresses an environment into a latent state and rolls it forward (Ha and Schmidhuber, 2018), a physics-informed network carries the equation error of a differential operator in its loss (Raissi et al., 2019), a generative diffusion model learns the score of a noised law and samples by reversing a stochastic differential equation (Ho et al., 2020; Song et al., 2021), and their native criterion is a held-out loss. IDS leaves these criteria in place and claims zero result about the families themselves. It asks a separate bookkeeping question: which reference defines relative information,

what does a declared observation channel preserve, and what does each finite reduction leave residual?

Uncertainty and information. Bayesian inference represents what is known about a law λ as a prior π and updates it with data $\mathcal{D}$ of likelihood $L_{\mathcal{D}}$ into a posterior $\mu_{\mathcal{D}} \propto L_{\mathcal{D}} \pi$ (Bayes, 1763), the law of least relative entropy to the prior under the evidence (Williams, 1980; Shore and Johnson, 1980; Zellner, 1988); Lindley (1956) measures the information an experiment provides as the relative entropy of the posterior from the prior, averaged over the outcome. Shannon's source of law p carries the entropy $H(X) = -\sum_x p(x) \log p(x)$, in nats; the mutual information $I(X; Y) = \sum_{x,y} p(x, y) \log \frac{p(x,y)}{p(x)p(y)}$ measures what an output tells about an input and stays equal or decreases under post-processing; the rate-distortion function $R(D) = \inf\{I(X; \hat{X}) : \mathbb{E}\|X - \hat{X}\|^2 \leq D\}$ is the least information a reconstruction of distortion at most D carries about the source (Shannon, 1948, 1959; Cover and Thomas, 2006).

Distinguishing and moving laws. The relative entropy of a law P from a law Q , $D(P \| Q) = \int \log \frac{dP}{dQ} dP \geq 0$, vanishes exactly when $P = Q$ and stays equal or decreases under every channel, with equality under a sufficient one (Kullback and Leibler, 1951; Csiszár, 1967); Blackwell (1953) orders experiments by garbling, and the projection of Csiszár (1975) selects, in a convex set, the law of least relative entropy to a reference. Between neighbouring laws of a smooth family, relative entropy is at second order a quadratic form, the Fisher information $D(p_{\theta} \| p_{\theta+d\theta}) = \frac{1}{2} g_{ij}(\theta) d\theta^i d\theta^j + o(\|d\theta\|^2)$, which Čencov (1982) characterized, up to a factor, as the metric invariant under sufficient Markov morphisms (Amari and Nagaoka, 2000; Ay et al., 2017). The quadratic Wasserstein distance between two probability measures is the least kinetic energy of a flow carrying one onto the other, $W_2^2(\mu_0, \mu_1) = \inf \int_0^1 \int \|v_t\|^2 d\mu_t dt$ under $\partial_t \mu_t + \nabla \cdot (\mu_t v_t) = 0$ (Benamou and Brenier, 2000; Villani, 2009); in that geometry the heat equation is the gradient flow of the entropy $\int \rho \log \rho$ (Jordan et al., 1998; Otto, 2001), the entropy decreases at the rate of the Fisher information of ρ , de Bruijn's identity (Stam, 1959), and the relative entropy to a reference whose Bakry–Émery curvature is bounded below by K is K -convex along displacement interpolation (Bakry and Émery, 1985; McCann, 1997; von Renesse and Sturm, 2005), with a squared metric slope equal to the relative Fisher information (Otto and Villani, 2000; Ambrosio et al., 2008).

Organization away from equilibrium: dissipative structures. A dissipative structure is an organization maintained away from equilibrium by fluxes exchanged with reservoirs (Nicolis and Prigogine, 1977). For the overdamped diffusions and the finite-state chains used below, equilibrium satisfies detailed balance and the stationary irreversible current vanishes; a connected structure embedded in reservoirs at distinct parameters can sustain stationary currents and produce entropy. Stochastic thermodynamics makes this quantitative: on a Markov process with rates k_{ij} and stationary law p the entropy production is $\sigma = \frac{1}{2} \sum_{ij} J_{ij} \mathcal{A}_{ij} \geq 0$ with $J_{ij} = p_i k_{ij} - p_j k_{ji}$ and $\mathcal{A}_{ij} = \log \frac{p_i k_{ij}}{p_j k_{ji}}$, current times affinity (Schnakenberg, 1976); on a path ω of a process of law P and reversed law $P^\dagger$ the entropy production is $\Sigma(\omega) = \log(dP/dP^\dagger)(\omega)$, whose exponential averages to one and whose mean is a relative entropy (Seifert, 2005, 2012; Esposito and Van den Broeck, 2010); local detailed balance ties the asymmetry of each transition to the entropy received by the reservoir that mediates it. A coarse description of the same process reads a share of that production, the remainder being carried by the finer level (Esposito and Van den Broeck, 2010; Gomez-Marin et al., 2008; Degünther et al., 2024).

Example: Brownian gyrator. A minimal, exactly solvable nonequilibrium heat-engine model (Filliger and Reimann, 2007) has two overdamped coordinates x_1, x_2 coupled by a spring of stiffness k , each tied to a wall by a spring k_1, k_2 and each immersed in its own bath at temperature T_1, T_2 :

$$dx_1 = [-k_1 x_1 - k(x_1 - x_2)] dt + \sqrt{2T_1} dW_1, \quad dx_2 = [-k_2 x_2 - k(x_2 - x_1)] dt + \sqrt{2T_2} dW_2.$$

Everything is Gaussian and in closed form: the stationary covariance solves a Lyapunov equation and, for $k > 0$ and $T_1 \neq T_2$, the stationary state carries a circulating probability current, heat

flows from the hot bath to the cold one through the spring and entropy is produced at a rate proportional to $k^2(T_1 - T_2)^2$; at $k = 0$ or at equal temperatures the current and the production vanish (Figure 1). The gyrator is the object on which the kernel is stress-tested; a driven ring and a metastable double well complete it where the gyrator is degenerate.

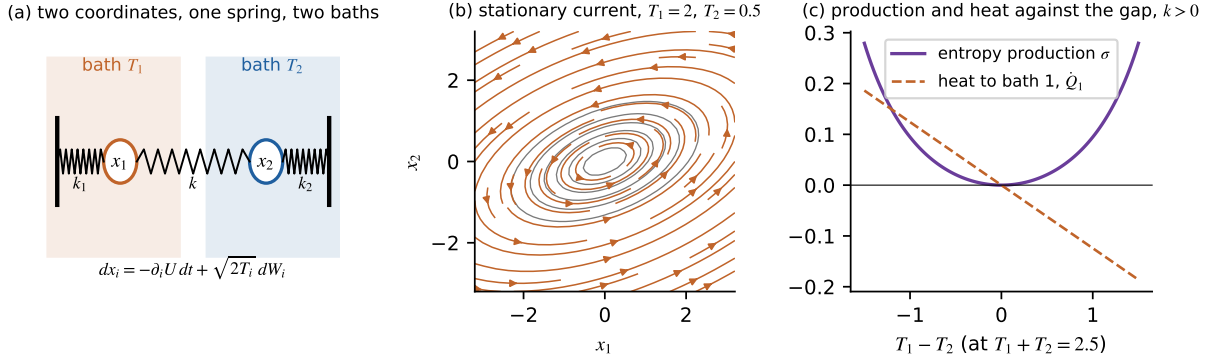


Figure 1: Brownian gyrator. (a) Two coordinates, one coupling spring, two walls, two baths. (b) The stationary probability current at $T_1 = 2$, $T_2 = 0.5$ (orange) and the level sets of the stationary density (grey). (c) The entropy production and the rate of heat delivered to the hot bath against the temperature gap, for $k > 0$: the stationary current and the entropy production stay away from zero exactly when $k(T_1 - T_2) \neq 0$.

What is missing. Most informational tools carry little physics: a graphical model has cliques and correlations, with the reservoirs and the balances of the phenomenon outside its description, and the geometry of hypotheses is that of a chosen parametrization, so that a change of observation channel and a change of coordinates are hard to tell apart. Most thermodynamic tools carry little belief: entropy production is a property of a path law, and the state of knowledge of the observer stays outside the bookkeeping. In practice the two sides are stacked: a sampler for the paths, a solver for the density, a coarse Markov model for the symbols, each under its own native criterion, with the joints between them open. Scientific models are finite descriptions of systems richer than any description, and the usual question is whether the description predicts well. We ask a second question: what information did the reduction leave residual, and can that part be carried consistently from the physical process to the geometry of the model and to the observer's belief?

Proposal. Informational Dissipative Structures (IDS) is our proposal: one framework in which the relative information $D(P \parallel R)$ of a law P from a declared reference R is the single primitive, read on three spaces, and in which a finite structure is accepted together with the ledger of what its resolution leaves residual. Every statement names its reference, and the informational specification is complete once its references are declared. One consequence organizes the paper: one bookkeeping reads phenomena of different natures and carries each of them, thermodynamically consistent, across the levels of a resolution tower, a notion made precise in Section 2, their coupling staying open (Q5), so that a driven diffusion, the cells a resolution cuts on it, the coarse Markov model those cells carry and the symbols read on them live on one sheet, their resolution-dependent rulers ordered by the channel order and their balances read in the same reference-explicit calculus (Figure 2). Three benchmarks of different natures carry the demonstration, a Brownian gyrator between two thermal baths, a driven ring with a work port and a metastable double well whose symbols close; the coupling of several such structures through a shared port is named as an open direction (Q5).

Outline of the construction. Informational Dissipative Structures (IDS) starts from a deliberately small datum: candidate processes, declared observation channels and declared references. A channel first determines which candidates are experimentally indistinguishable. On a smooth stratum of the identifiable quotient, relative entropy then provides the local Fisher ruler. Probability measures over that stratum describe the remaining uncertainty, and the Wasserstein slope of relative entropy supplies a calibrated length for revising those beliefs. A finite structure

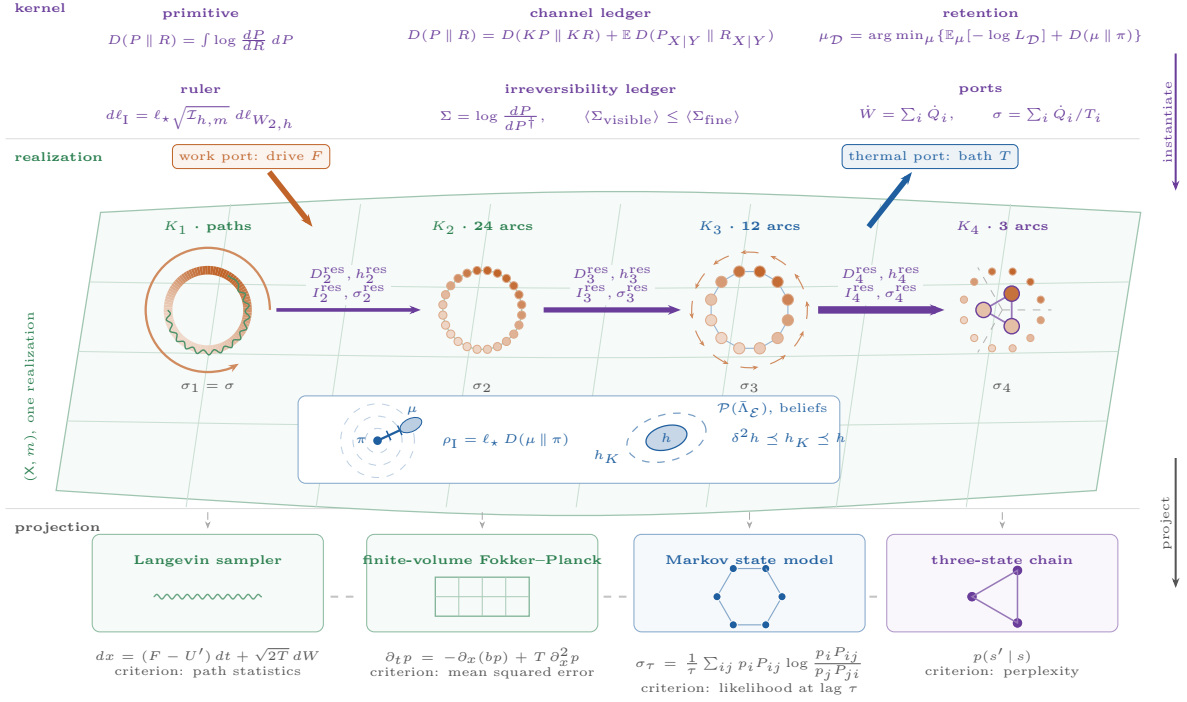


Figure 2: Driven ring at four nested resolutions. Top, the kernel: the primitive, the channel ledger, the retention rule, the ruler, the irreversibility ledger and the port balance. Middle, one realization: the state space of the driven ring with its reference measure, read at four ordered resolutions, the fine path description and the partitions into twenty-four, twelve and three arcs. The shade of a cell averages the local entropy production that the geometry carries there, on one scale for the four levels, so the shade of a coarse cell is the mean of the shades of the fine cells it covers; it stays distinct from σ_N , the production that the coarse level itself reads. Between levels, an arrow carries the ledgers that the coarser level leaves residual, in relative information, in Fisher information, in mutual information and in production, its width growing with the residual share; σ_N is the production that the level reads. The drive enters through a work port and the bath receives through a thermal port, the two arrows drawn at one width, as the stationary balance requires. The inset holds the third space: a prior, a belief, the radial geodesic between them with one mark per nat on the level sets of the calibrated radius, and two nested precision ellipses, the fine one inside the coarse one, within the bounds of the stability theorem. Bottom, the projections: the model that each level induces, a Langevin sampler, a finite-volume Fokker–Planck solver, a Markov state model at lag τ and a three-state chain, each under its own native criterion, with the joints between them drawn open. In the margin, the two operations, instantiate and project.

enters afterwards, as one more channel in a resolution tower, together with the exact ledger of what it leaves residual. The chain of the paper is

process $\rightarrow$ channel $\rightarrow$ identifiable law $\rightarrow$ geometry $\rightarrow$ resolution $\rightarrow$ finite structure,

and the kernel is free of any graph or clique structure (Figure 2). A model is a channel-relative geometry; a finite structure is accepted together with the ledger of what its resolution leaves residual.

Contributions. The ingredients are established; the contribution is their assembly into one reference-explicit calculus of reduction, organized around five pillars and two consequences. The thirteen numbered statements of the body are named here, with their identifier, their status and their section.

- **A, informational dissipative structures.** An informational dissipative structure is a finite-resolution realization whose coarse state closes predictively, whose path law is irreversible, and whose irreversibility is sustained by declared physical ports with closed

balances. The irreversibility seen by a time-reversal-compatible quotient stays at or below that of the fine process and is recovered by an exhaustive compatible refinement. One bookkeeping reads phenomena of different natures and carries them through a resolution tower; the benchmarks below test the clauses on complementary systems.

- T13 · Irreversibility Ledger Theorem (K5, with its corollary R6) · proved · Section 5
- Informational Dissipative Structure (H1) and its ports (H2) · definition; measured on the gyrator, the driven ring and the double well · Section 7
- **B, the channel ledger.** Every channel splits relative information into a visible and a residual part, exactly, and the ledger characterizes relative entropy.
 - T1 · Declared-Reference Principle (K0) · principle · Section 3
 - T2 · Channel Ledger Theorem (K2, with C1) · imported · Section 3
 - T3 · Retention Theorem (K3) · imported, proof recalled · Section 3
 - T4 · Recovery-Kernel Theorem (K4) · imported · Section 3
- **C, the differential ledger.** On the quotient of identifiable laws the second-order differential of the ledger is the Fisher metric of the observation, and the missing Fisher information is the expected conditional variance of the score.
 - T5 · Operational Fisher Theorem (O1) · imported · Section 4
 - T6 · Missing-Fisher Ledger Theorem (O3) · proved · Section 4
- **D, the slope geometry.** On a length space the metric slope of a lower semicontinuous functional is a ruler with a floor; for relative entropy under Wasserstein transport the ruler is the Fisher functional, certainty lies on a horizon, the weight is pointwise rigid and the prior-centred distance is relative entropy itself.
 - T7 · Slope-Geometry Proposition · proved · Section 4
 - T8 · RESG Theorem: Entropy Floor, Constant Slope, Radial Identity (G3, G4, G5) · proved · Section 4
- **E, resolution recovery.** Along an exhaustive tower of channels, relative information, Fisher information, Bayesian information and calibrated radius recover their fine values monotonically.
 - T9 · Resolution Ledger Theorem (R1, R2) · proved · Section 5
 - T10 · Resolution Recovery Theorem (R5) · proved · Section 5
- **Two consequences.** A δ -almost sufficient projection is a δ -bi-Lipschitz map of the slope geometry, and a finite description obeys an exact predictive ledger, $I_{\text{pred}} = I_{\text{ret}} + I_{\text{res}}$, which defines its closure.
 - T11 · Almost-Sufficient Stability Theorem (G10, with its corollary G11) · proved · Section 5
 - T12 · Predictive Closure Ledger Theorem (E6) · proved · Section 6

Organization. A statement lives at one of two levels, printed after its identifier: the *kernel* is what follows from the declared datum in the language of relative information alone; a *realization* instantiates the kernel on a named space, where statements gain their closed forms; a *projection*, a resolution of a realization, keeps the map to the realization and the ledger of what it leaves residual, and is stated at the level of the realization it resolves. A numerical check on a benchmark is cited as a gate, such as X1.1; every gate was declared in writing before its first computation, and Appendix S.D names every one of them. Each statement prints its level and its status: IMPORTED names a standard theorem with its reference, PROVED a proof in this paper or its supplement, MEASURED a fact established on a realization by gates, PRINCIPLE and DEFINITION what they say. Section 2 fixes the primitive, the three spaces and what is primitive; Section 3 states the formal core; Section 4 the two differentials of the primitive; Section 5 resolution as an ordered family of channels; Section 6 the emergent finite structure; Section 7 the physical realization and the numerical experiments; Section 8 the relation to prior work, the scope and the limitations; Section 9 concludes. The thirteen numbered statements T1 to T13 carry the body; every other

statement is stated in the supplement (Appendix S.C).

2 Proposed formulation

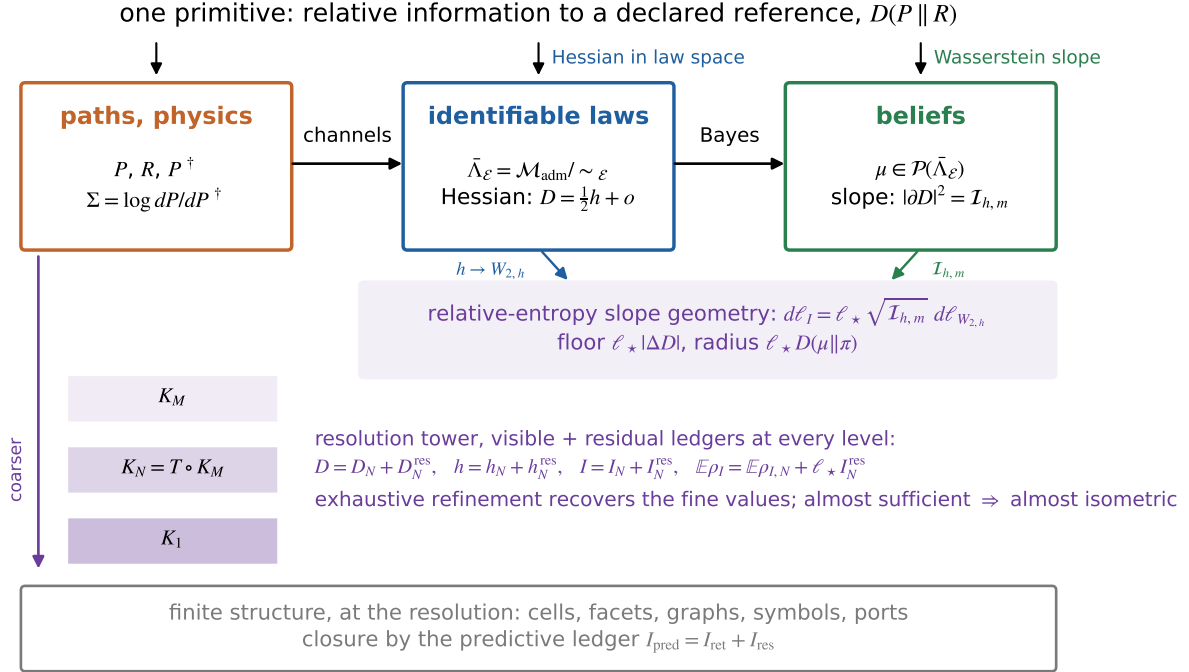


Figure 3: Primitive, three spaces, two differentials. One primitive, relative information to a declared reference, read on three spaces and differentiated twice: the Hessian in the space of identifiable laws gives the Fisher metric h , the Wasserstein slope on beliefs gives the Fisher functional $\mathcal{I}_{h,m}$, and the two meet in the relative-entropy slope geometry; a tower of channels carries the visible and residual ledgers down to cells, graphs and symbols.

Problem. Describe, from relative information to declared references, the laws an observer can tell apart, the beliefs over those laws and the cost of revising them, the information that every finite description leaves residual, and the structure that a driven physical process maintains, in one architecture whose derived and realization-level claims are checked on explicit benchmarks.

The relative information of a law P from a declared reference R ,

$$D(P \parallel R) = \int \log \frac{dP}{dR} dP \geq 0, \quad (2.1)$$

is the primitive. A problem is a datum $\mathfrak{J} = (\mathcal{M}_{\text{adm}}, \mathcal{S}, \mathcal{E}, R, \pi; m_{\Lambda})$: the admissible candidates, their semantics, the operational access $\mathcal{E}$ (the declared experiments), the path reference R , the prior π and, when the geometry is invoked, the calibration measure m_{Λ} on a stratum. Within IDS, the informational specification is complete once its references are declared. Here R and π are probability references, so $D(P \parallel R) \geq 0$ and $D(\mu \parallel \pi) \geq 0$; the calibration measure m_{Λ} of the geometry may be σ -finite, Lebesgue measure among the cases used below, and the functional $\mathcal{H}_m(\mu) = \int \log(d\mu/dm_{\Lambda}) d\mu$ it defines then takes values of both signs, the prior-centred radial statements keeping the probability reference π . The primitive is read on three spaces and differentiated twice (Figure 3, Table 1).

Two objects share a symbol in much of the literature and are typed apart here: the ensemble law $p_t \in \mathcal{P}(\mathbf{X})$ of a physical state, read against a reference $m_{\mathbf{X}}$ on the state space, and a belief $\mu \in \mathcal{P}(\tilde{\Lambda}_{\mathcal{E}})$ over identifiable laws, read against π or m_{Λ} . The slope construction of Section 4 is

space	object	declared reference	channel	reading of D
physical paths	path law P	the reversal $P^\dagger$, or another declared path reference R	observation of paths	irreversibility against $P^\dagger$, relative path-law cost against R (K5)
identifiable laws	$K_\lambda, \lambda \in \bar{\Lambda}_\mathcal{E}$	the law of a neighbouring class	experimental channel K	distinguishability and its ledger (K2, O1)
beliefs on laws	$\mu \in \mathcal{P}(\bar{\Lambda}_\mathcal{E})$	prior π , calibration m_Λ	Bayesian experiment	update, information geometry (K3, G3)

Table 1: One primitive read on three spaces, each with its declared reference and its channel.

abstract before it is epistemic: in the informational reading it acts on beliefs over identifiable laws; in the physical benchmarks the same construction is instantiated on ensemble laws over a state space. The two uses share the calculus and differ in semantics. On beliefs the reading is a geometry: the transport distance built on the law ruler h , weighted by the slope of relative entropy,

$$d\ell_1 = \ell_\star \sqrt{\mathcal{I}_{h,m}} d\ell_{W_{2,h}}$$

the *relative-entropy slope geometry* (RESG, T8, Section 4.2), where $\ell_\star$ is the length of one nat, $\mathcal{I}_{h,m}$ the relative Fisher information of a belief and $W_{2,h}$ the quadratic transport distance built on h (Section 4.2); a finite description is a channel in a tower, each level carrying its *resolution ledger* (T9, Section 5).

Reading on paths. On the paths of a process the primitive reads irreversibility: when a reversal $P^\dagger$ equivalent to P exists,

$$\Sigma = \log \frac{dP}{dP^\dagger}, \quad \mathbb{E}_P[e^{-\Sigma}] = 1, \quad \mathbb{E}_P[\Sigma] = D(P \parallel P^\dagger) \geq 0, \quad (2.2)$$

and every quotient of the paths that commutes with time reversal reads at most the fine irreversibility, $\langle \Sigma_{\text{visible}} \rangle = D(KP \parallel (KP)^\dagger) \leq \langle \Sigma_{\text{fine}} \rangle$ (K5, a consequence of the ledger of Section 3; Kawai et al., 2007; Gomez-Marin et al., 2008). On the gyrator every scalar linear observable is a stationary Gaussian process, hence reversible: an observer of one coordinate reads zero arrow at every tested lag, while an observer of both approaches σ as its time resolution is refined (Section 7).

Thermodynamic consistency. A description is *thermodynamically consistent* when its three ledgers close together: the channel ledger of relative information, visible plus residual (T2); the irreversibility ledger of its path law, Equation (2.2), whose visible share obeys $\langle \Sigma_{\text{visible}} \rangle \leq \langle \Sigma_{\text{fine}} \rangle$ under every quotient that commutes with time reversal (T13); and the port ledger of its realization, $\dot{W} = \sum_i \dot{Q}_i$ and $\sigma = \sum_i \dot{Q}_i / T_i$ in the stationary state (H2). A tower of descriptions is thermodynamically consistent *across scales* when every level closes its port balances and the informational and irreversibility ledgers obey their refinement laws: their visible shares grow along the order and reach the fine values on an exhaustive tower (T10, R6). A *scale* is a level of that tower. A general composition law for the ports themselves under an arbitrary coarse graining stays open (Q3, Q5).

What is primitive. Table 2 says in one glance what is declared, what is derived and what belongs to a realization.

object	primitive	derived	realization-dependent
relative information to a declared reference	yes		
observation channel, operational access $\mathcal{E}$	declared		
identifiable quotient $\bar{\Lambda}_{\mathcal{E}}$		yes (K1)	
Fisher metric h		yes (O1)	
Bayesian posterior		yes, under retention (K3)	
relative-entropy slope geometry		yes, under HR0 (G3)	
resolution tower	declared, or-ordered		
harmonic modes			optional (R3)
cells, graphs, symbols			finite resolution (E2, E3, E5)
cliques, factorizations			generator realization (E3)
ports, temperature, energy			thermodynamic typing (H2, K6)

Table 2: What is primitive, what is derived, what belongs to a realization.

3 Formal core

3.1 Declared data

Four definitions fix the objects that the principles use; the notation follows, and every statement carries a status.

Definition D1 (Processes and paths · kernel · DEFINITION). A *process* is a probability law on the paths of a state space $\mathbf{X}$, with the family of its finite-dimensional laws. Its *path law* is written P ; its *reversal* $P^\dagger$ is the law of the time-reversed paths, a probability law equivalent to P when it exists. Its ensemble law at time t is $p_t \in \mathcal{P}(\mathbf{X})$. Information is counted in nats.

Definition D2 (Observable laws and strata · kernel · DEFINITION). Under an experiment $e = (a, K)$ of the operational access $\mathcal{E}$, an action and a channel, the *observable law* of a candidate M is $p_M(\cdot \mid e)$. For a class λ of the identifiable quotient $\bar{\Lambda}_{\mathcal{E}}$ (K1), K_λ denotes the observable law of the class under the channel K , the same for every M in the class. A *stratum* is a smooth part of $\bar{\Lambda}_{\mathcal{E}}$, a *regular stratum* a stratum on which the signature is differentiable; h is the Fisher metric of the observable laws on a regular stratum (O1), and $\preceq$ the order of quadratic forms.

Definition D3 (References and the datum · kernel · DEFINITION). Every informational quantity of the kernel is referenced explicitly, and the relative information $D(P \parallel R)$ of a law P from a declared reference is the primitive from which the differential and geometric quantities are derived. Three reference measures are declared, the *path reference* R of a process, the *epistemic reference* π , the prior on the identifiable quotient, and, when the geometry is invoked, the *calibration measure* m_Λ on a stratum; the operational access $\mathcal{E}$ is a family of experiments, declared beside them. The datum is $\mathfrak{J} = (\mathcal{M}_{\text{adm}}, \mathbf{S}, \mathcal{E}, R, \pi; m_\Lambda)$, and zero internal structure is imposed on a candidate M .

Definition D4 (Beliefs and information · kernel · DEFINITION). A *belief* is a probability measure $\mu \in \mathcal{P}(\bar{\Lambda}_{\mathcal{E}})$ on the identifiable quotient. The *random law* Λ is a random element of $\bar{\Lambda}_{\mathcal{E}}$ distributed as the prior π ; for the outcome Y of an experiment, π_Y is the posterior after Y , and $I(\Lambda; Y)$ the mutual information between the law and the outcome, equal to $\mathbb{E}_Y D(\pi_Y \parallel \pi)$ (B1).

Notation. $\log$ is the natural logarithm; $\mathbb{E}_P$ is the expectation under P , written $\mathbb{E}$ when the law is clear; $\mathcal{P}(\cdot)$ is the set of probability measures on a space; $K : \mathbf{X} \rightsquigarrow \mathbf{Y}$ is a Markov kernel, a channel, and KP the image of a law P by K ; ∇_h , $\|\cdot\|_h$, Δ_h , div_h are the gradient, norm, Laplacian, and divergence.

Laplace–Beltrami operator and divergence of h ; $\operatorname{div}_m Y = e^V \operatorname{div}_h(e^{-V} Y)$ for $m = e^{-V} dV_h$; $\mathbf{D}$ is the diffusion matrix of a diffusion, $\mathcal{D}$ a data set; visible and residual parts of a ledger at resolution N carry the superscripts *vis* and *res*. An f -divergence is $\int f(dP/dQ) dQ$ for a convex f with $f(1) = 0$ (Csiszár, 1967).

3.2 Channel and quotient

Postulate K0 (T1, Declared-Reference Principle · kernel · PRINCIPLE). Every information is relative to a reference declared on the space where it is computed. The path reference R , the prior π and the calibration m_Λ stay distinct, and an ensemble law on $\mathbf{X}$ is read against its own reference $m_\mathbf{X}$.

Definition K1 (Channels, signature and identifiable quotient · kernel · DEFINITION). Every observation, reduction or resolution is a channel. The signature of M under $\mathcal{E}$ is $F_\mathcal{E}(M) = \{p_M(\cdot | e)\}_{e \in \mathcal{E}}$; two candidates are equivalent when their signatures coincide,

$$M \sim_\mathcal{E} M' \iff F_\mathcal{E}(M) = F_\mathcal{E}(M'), \quad \bar{\Lambda}_\mathcal{E} = \mathcal{M}_{\text{adm}} / \sim_\mathcal{E}, \quad (3.1)$$

and $\bar{\Lambda}_\mathcal{E}$ is the space of the laws that the experiments tell apart. On the gyrator with $\theta = (k_1, k_2, k, T_1, T_2)$, a window of one observed coordinate leaves four identifiable combinations of the five parameters, two coordinates leave five; at $k = 0$ one coordinate sees two, the stiffness $k_1 + k$ and T_1 , and its null space holds $(1, 0, -1, 0, 0)$, e_{k_2} and e_{T_2} (gates X1.1, X1.4b).

Theorem K2 (T2, Channel Ledger Theorem: the exact ledger and data processing · kernel · IMPORTED). For every channel K ,

$$D(P \| R) = D(KP \| KR) + \mathbb{E}_{KP} D(P_{X|Y} \| R_{X|Y}), \quad (3.2)$$

on standard Borel spaces, the visible part plus the residual part carried by the conditionals. Residual names the exact part of the fine relative information that the current channel leaves aside; it is a complement inside an identity, and it stays free of any claim about a latent variable. Hence $D(KP \| KR) \leq D(P \| R)$, and when $D(P \| R) < \infty$ equality holds exactly when the channel is sufficient for the pair (Kullback and Leibler, 1951, Thm 4.1), (Csiszár, 1975; Cover and Thomas, 2006).

Corollary C1 (The ledger characterizes relative entropy, finite alphabets · kernel · IMPORTED). Let I assign a real number to each pair (p, r) of laws on finitely many points with $p \ll r$. Suppose I is measurable in r at fixed p , invariant under a common relabelling of both laws, vanishes on (p, p) , and obeys the exact ledger of every partition channel (Theorem K2), $I(w \circ (p^1, \dots, p^n) \| \tilde{w} \circ (\tilde{p}^1, \dots, \tilde{p}^n)) = I(w \| \tilde{w}) + \sum_i w_i I(p^i \| \tilde{p}^i)$, the visible part plus the residual part averaged under the visible law. Then $I = c D(\cdot \| \cdot)$ for a constant c (Leinster, 2019, Thm 2.1); the unit of one nat fixes $c = 1$, and the functorial form of the statement is Baez and Fritz (2014), extended to standard Borel spaces by Gagné and Panangaden (2018); the characterization is a finite-alphabet statement, and the kernel’s use of the ledger on general spaces rests on Equation (3.2) itself. Negative controls: on fifty random joints on 3×3 states the discrepancy of the ledger is $2 \cdot 10^{-16}$ for relative entropy and reaches 0.05 for the squared Hellinger distance, 0.37 for the total variation and 10.8 for χ^2 ; the telescoping over two channels behaves the same way (gates C1.1, C1.2).

3.3 Retention

Theorem K3 (T3, Retention Theorem: information projection and Bayes · kernel · IMPORTED). Under hard constraints, wherever the information projection onto the declared constraint set exists, the retained law is that projection (Csiszár, 1975). Under likelihood evidence L with $Z = \int L d\pi \in (0, \infty)$, the least-deformation rule from the prior is Bayes,

$$\mu_{\mathcal{D}} = \arg \min_{\mu} \left\{ \mathbb{E}_{\mu}[-\log L] + D(\mu \| \pi) \right\} = \pi_{\mathcal{D}} := \frac{L \pi}{Z}, \quad (3.3)$$

Bayes' rule in its Gibbs variational form (Zellner, 1988).

Proof. For every belief μ , $\mathbb{E}_\mu[-\log L] + D(\mu \parallel \pi) = \int \log \frac{d\mu}{L d\pi} d\mu = \int \log \frac{d\mu}{Z d\pi_{\mathcal{D}}} d\mu = D(\mu \parallel \pi_{\mathcal{D}}) - \log Z \geq -\log Z$, and relative entropy vanishes exactly on equal laws (gate C2.3). On finite simplices, for priors of full support and likelihoods with positive entries, relative entropy is the differentiable f -divergence whose least-deformation update coincides with $\pi_{\mathcal{D}}$ for every such prior and likelihood (C2, proved in the supplement, Appendix S.B); Bissiri and Walker (2012) and Bissiri et al. (2016) reach the same selection through a coherence requirement. $\square$

Corollary B1 (Belief and the information of an experiment, Lindley · kernel · IMPORTED). *The expected information of an experiment is the mean realized relative information of its posterior, $I(\Lambda; Y) = \mathbb{E}_Y D(\pi_Y \parallel \pi)$ (Lindley, 1956); post-processing contracts it, by Equation (3.2) applied to the joint law against the product (Theorem K2).*

3.4 Sufficiency

Theorem K4 (T4, Recovery-Kernel Theorem: sufficiency and recovery · kernel · IMPORTED). *When $D(P \parallel R) < \infty$, a zero residual term in the ledger Equation (3.2) of a channel K is sufficiency (Kullback and Leibler, 1951, Thm 4.1). For a dominated declared family $\{P_\lambda\}$ on standard Borel spaces, sufficiency of K for the augmented family $\{R\} \cup \{P_\lambda\}$ is equivalent to the existence of a common recovery Markov kernel L with $L K R = R$ and $L K P_\lambda = P_\lambda$ for every λ (Blackwell, 1953; Petz, 1988), and a sufficient channel preserves the Fisher metric of a regular stratum.*

4 Information calculus and relative-entropy slope geometry

The primitive is differentiated twice, and Fisher information appears twice, as two objects: the Hessian of D in the space of identifiable laws is a tensor h on the stratum; the Wasserstein slope of D on measures over the stratum is a scalar functional $\mathcal{I}_{h,m}$ on beliefs. The two meet in the relative-entropy slope geometry (Figure 4).

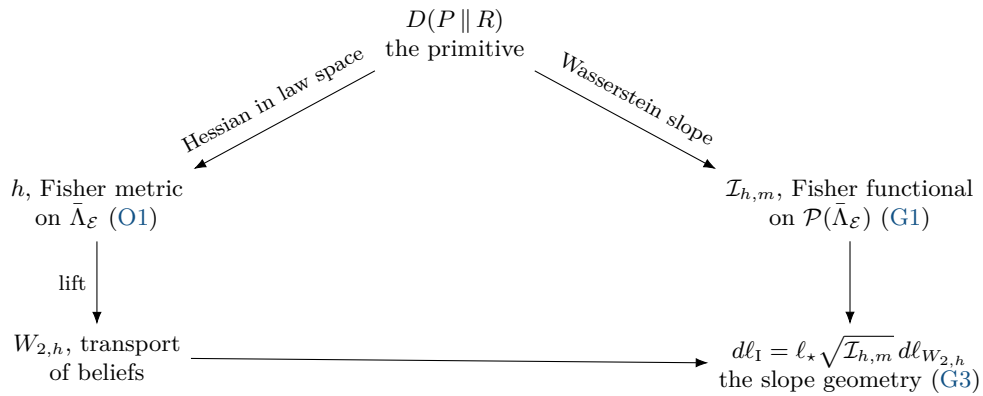


Figure 4: Two differentials of one primitive. The Hessian gives the tensor h on identifiable laws; the slope gives the functional $\mathcal{I}_{h,m}$ on beliefs; the transport distance built on h , weighted by the slope, is the calibrated length.

4.1 Differential in law space

Theorem O1 (T5, Operational Fisher Theorem · kernel · IMPORTED). *On a regular stratum of the quotient,*

$$D(K_\lambda \parallel K_{\lambda+d\lambda}) = \frac{1}{2} h_K(d\lambda, d\lambda) + o(\|d\lambda\|^2), \quad h_{T \circ K} \preceq h_K. \quad (4.1)$$

The metric is induced by the observational access; it contracts under every channel and stays under a sufficient one; up to a factor it is the metric invariant under sufficient Markov morphisms, on finite sample spaces by Čencov (1982) and on parametrized measure models by Ay et al. (2017).

Theorem O3 (T6, Missing-Fisher Ledger Theorem: score projection and residual Fisher · kernel · PROVED). For a family differentiable in quadratic mean with scores $S_v \in L^2$ and an observation $Y = K(X)$ through a Markov channel K free of the parameter, the observable score is the conditional expectation of the fine score, $S_v^Y = \mathbb{E}[S_v | Y]$ (O2), and

$$h = h_Y + h_Y^{\text{res}}, \quad h_Y^{\text{res}}(v, v) = \mathbb{E} \text{Var}(S_v | Y) \geq 0, \quad (4.2)$$

the residual Fisher being the Hessian of the residual relative information of Equation (3.2). Along the null direction of the observable Fisher all of the fine Fisher is residual (gates X1.2, X1.3b, X2.2; Figure 5).

The identities are classical: Fisher’s identity for the score of an observation, the missing-information principle of Orchard and Woodbury (1972), Louis’s formula for the observed information (Louis, 1982), and the fraction of missing information of Meng and Rubin (1991). IDS reads them as the second-order differential of the exact channel ledger and places them in the same resolution tower as relative information, mutual information and calibrated radius; the residual share of the Fisher information along a direction is the fraction of missing information, and a δ -almost sufficient projection is one whose largest fraction of missing information is at most $1 - \delta^2$ (G10).

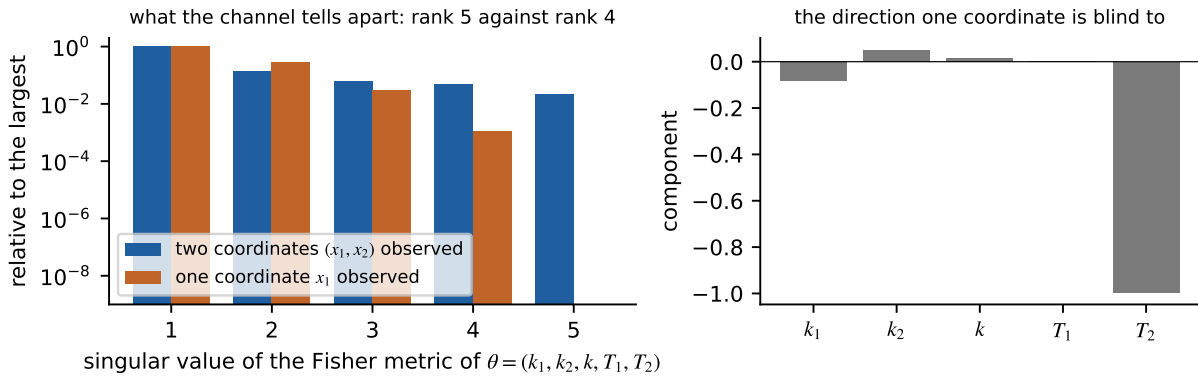


Figure 5: Fisher rank under two observation windows. Left: the singular values of the Fisher metric of $\theta = (k_1, k_2, k, T_1, T_2)$ from a window of the gyrator’s trajectory, 64 samples at step 0.2 without observation noise, two coordinates observed (blue) and one coordinate observed (orange): rank five against rank four. Right: the components of the direction that one coordinate is blind to; along it the whole Fisher information is residual.

4.2 Relative-entropy slope geometry

RESG extends the belief-cost geometry of Caraffa (2026), whose reference is the volume, to a declared reference.

Proposition S (T7, Slope-Geometry Proposition · kernel · PROVED). Let (X, d) be a length space, $H : X \rightarrow (-\infty, +\infty]$ lower semicontinuous, $g = |\partial H|_d$ a strong upper gradient of H (Ambrosio et al., 2008, Def. 1.2.1) and $\ell_\star > 0$ a constant. For $L_H(\gamma) = \ell_\star \int g(\gamma_t) |\dot{\gamma}_t| dt$, every absolutely continuous curve with $L_H(\gamma) < \infty$ satisfies $L_H(\gamma) \geq \ell_\star |H(\gamma_1) - H(\gamma_0)|$; and when a curve of maximal slope from x , with $-\frac{d}{dt}H = g^2 = |\dot{x}|^2$, converges to a minimizer $x_\star$ of H with $H(x_t) \rightarrow H(x_\star)$, in finite time or as $t \rightarrow \infty$, the least length from x to $x_\star$ is $\ell_\star [H(x) - H(x_\star)]$, the infimum running over the curves that reach $x_\star$ in the value of H , defined on $[0, 1)$ with $H(\gamma_t) \rightarrow H(x_\star)$.

Proof. The strong upper gradient gives $|H(\gamma_1) - H(\gamma_0)| \leq \int g(\gamma_t) |\dot{\gamma}_t| dt$ (Ambrosio et al., 2008, Def. 1.2.1); multiply by $\ell_\star$. Along the curve of maximal slope (Ambrosio et al., 2008, Def. 1.3.2), $\ell_\star \int_0^T g(x_t) |\dot{x}_t| dt = \ell_\star \int_0^T (-\frac{d}{dt} H(x_t)) dt = \ell_\star [H(x) - H(x_T)]$, which tends to $\ell_\star [H(x) - H(x_\star)]$; the first part makes this length the least one. $\square$

The construction is abstract before it is epistemic. Its epistemic realization is $X = \mathcal{P}_2(\bar{\Lambda}_\mathcal{E})$, $d = W_{2,h}$, $H = D(\cdot \| m_\Lambda)$: the points are beliefs over identifiable laws. Its physical realization is $X = \mathcal{P}_2(\mathbf{X})$, $d = W_{2,g_X}$, $H = D(\cdot \| m_X)$: the points are ensemble laws over a state space, and the gyrator benchmarks it. Both are invoked under one contract.

Contract HR0 (Geometric contract $\cdot$ kernel $\cdot$ DEFINITION). The stratum (Λ, h) is a complete Riemannian manifold; the reference is $m = w dV_h$ with $\text{Ric}_h + \text{Hess}_h(-\log w) \geq Kh$ for some real K (Bakry–Émery); beliefs are regular. Under HR0 the metric slope of $\mathcal{H}_m(\mu) = \int f \log f dm$, $f = d\mu/dm$, in $W_{2,h}$ is $|\partial \mathcal{H}_m|^2 = \mathcal{I}_{h,m} = \int \|\nabla_h \log f\|_h^2 d\mu$, its gradient flow is the Fokker–Planck equation and $\mathcal{H}_m$ is K -convex along displacement interpolation (G1, imported in Appendix S.B); for a normalized probability reference and $K > 0$ a logarithmic Sobolev inequality holds and gives exponential decay and Talagrand’s inequality (Bakry and Émery, 1985; Otto and Villani, 2000). The same contract, written for the state space $(\mathbf{X}, g_X, m_X)$ with g_X its Euclidean metric, covers the physical realization.

Definition G2 (The calibrated length $\cdot$ kernel $\cdot$ DEFINITION).

$$\boxed{d\ell_1 = \ell_\star \sqrt{\mathcal{I}_{h,m}} d\ell_{W_{2,h}}} \quad (4.3)$$

the slope of relative entropy is the ruler, $\ell_\star$ is a gauge, the length of one nat; L_1 is the length of a curve and d_1 the induced length pseudometric, symmetric with the triangle inequality under HR0 alone and a metric wherever points are separated, which the logarithmic Sobolev inequality guarantees on the extended space (Appendix S.B.2); the name calibrated distance is kept for it.

Theorem G3 (T8, RESG Theorem: Entropy Floor, Constant Slope, Radial Identity $\cdot$ kernel $\cdot$ PROVED).
Under HR0:

- (i) floor and horizon: every path pays at least $\ell_\star$ times the entropy difference of its endpoints, $d_1(\mu_0, \mu_1) \geq \ell_\star |\mathcal{H}_m(\mu_1) - \mathcal{H}_m(\mu_0)|$; the states of infinite relative entropy, exact certainty among them, are approached only at diverging calibrated distance and at finite transport distance: $W_{2,h}(\mu_n, \delta_\lambda) \rightarrow 0$ with $m(\{\lambda\}) = 0$ forces $d_1(\mu_0, \mu_n) \rightarrow \infty$;
- (ii) pointwise rigidity: for a scalar-conformal deformation of $W_{2,h}$ by a relief a , the slope of relative entropy at a regular belief with $0 < \mathcal{I}_{h,m} < \infty$, where a is continuous, is $\sqrt{\mathcal{I}_{h,m}}/a$, so it equals $1/\ell_\star$ there exactly when $a = \ell_\star \sqrt{\mathcal{I}_{h,m}}$ there; the weight $\ell_\star \sqrt{\mathcal{I}_{h,m}}$ gives the constant slope $1/\ell_\star$ at every such belief (G4);
- (iii) radial identity: with the prior as reference and an ergodic gradient flow, $\rho_1(\mu) = \ell_\star D(\mu \| \pi)$, relative entropy is the radial coordinate and the gradient flow is the radial geodesic (G5).

On the gyrator, in the physical realization with $(k_1, k_2, k) = (1, 0.7, 0.6)$: cooled to $T \rightarrow 0$ against the reference at $T = 1$, D grows like $\log(1/T)$ with slope 0.998 while W_2 stays below 1.30 (gate X3.3); the relaxation runs at equal temperatures $T_1 = T_2 = 1$ from the displaced stationary law $\mathcal{N}(m_0, C)$, with C the stationary covariance at $T_1 = 2$, $T_2 = 0.5$ and the offset $m_0 = (1.5, -1)$, towards the Gibbs law at $T = 1$, with the gauge $\ell_\star = \varepsilon/2 = 1$ of G8 ($\varepsilon = 2T$); on it the bare transport length is 1.857 and the length with weight $\mathcal{I}_{h,m}$ in place of $\sqrt{\mathcal{I}_{h,m}}$ is 9.15 against a drop of relative information of 3.554 (gate X3.4); with isotropic diffusion and a symmetric drift the relaxation is the gradient flow of $D(\cdot \| \pi)$, hence radial: its calibrated length is 3.55407, the drop of relative information 3.55402 and the entropy production measured on $2 \cdot 10^4$ paths 3.535 ± 0.023 (gates X3.1, X3.2, LIB.8). The proofs are in Appendix S.B; Figure 6 shows the Gaussian leaf under the two calibration references.

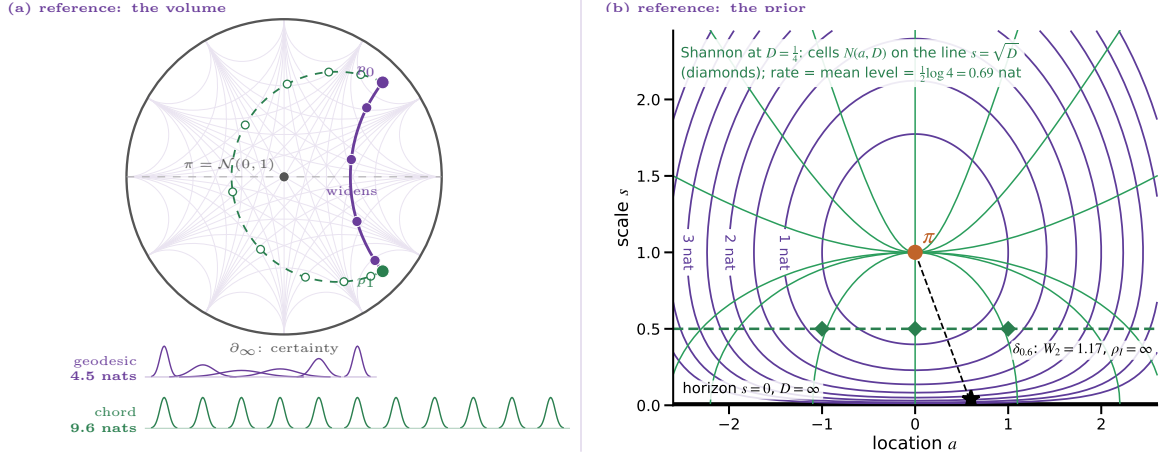


Figure 6: One leaf, two calibration references. (a) The volume as reference, drawn in the Poincaré disk of the Gaussian leaf: the geometry is hyperbolic and homogeneous, the prior $\pi = \mathcal{N}(0, 1)$ at the centre carries zero privilege and the certainties are the boundary; two beliefs, the geodesic between them, solid with one mark per nat, and the chord at fixed width, dashed; below, the belief along each route, one snapshot per nat of cost. (b) The prior as reference, drawn in the half-plane (a, s) of the same leaf: the geometry acquires a centre. Purple: the level sets of $D(\cdot \| \pi)$, the calibrated circles about π , half a nat apart. Green: the traces of the gradient flow, the radial geodesics. The horizon $s = 0$ is at infinite calibrated distance and at finite transport distance, the Dirac mass $\delta_{0.6}$ at $W_2 = 1.17$. The dashed line $s = \sqrt{D}$ carries the cells $\mathcal{N}(a, D)$ of Shannon’s channel at distortion $D = \frac{1}{4}$: their transport radius to the code point is s , their calibrated radius is the level they touch, and the mean level along the line under $a \sim \mathcal{N}(0, 1 - D)$ is the rate $\frac{1}{2} \log(1/D) = 0.69$ nat (gate S1.1).

Corollary G7 (Scale invariance · kernel · PROVED). *Rescaling the metric of Definition G2, $h' = \alpha h$ with the same reference measure, or with the volume of h' when the reference is the volume of h , leaves the calibrated length invariant (cf. Caraffa, 2026, Thm 4.23); n repetitions of an experiment turn h into nh , and the calibrated length is invariant under that constant rescaling for the stated reference choice.*

Corollary G8 (Diffusive-Unit Corollary · kernel · PROVED). *Under the diffusive-reference hypotheses of Appendix S.B.2, for a noise ε and the gauge $\ell_\star = \varepsilon/2$, the Girsanov relative entropy of a controlled path measure P against the reference R , completed by the entropy boundary term, bounds the calibrated length of its marginals (Definition G2), $\varepsilon D(P \| R) - \frac{\varepsilon}{2} [\mathcal{H}_m(\mu_1) - \mathcal{H}_m(\mu_0)] \geq L_1(\mu)$, with equality along the reference flow; $\varepsilon/2$ is the largest gauge for which this bound holds, and the energy form of the entropic cost behind it is that of Gentil et al. (2017, Cor. 5.3) and Chen et al. (2016): a path-space information price for diffusive revision. A reading of that price as heat requires a thermal realization under local detailed balance (H2).*

Corollary G9 (Shannon as posterior geometry · kernel · PROVED). *Under the radial identity of Theorem G3, $\ell_\star$ times the mutual information is the mean calibrated radius of the posteriors, and $\ell_\star$ times the rate–distortion function is the least mean calibrated radius of the posterior cells of a test channel under a mean squared transport radius to the code points; for a source without atoms zero distortion is the horizon, $R(0) = \infty$, while a finite alphabet gives $R(0) = H(X)$ (Shannon, 1959). For the Gaussian source the mean radius of the cells equals $\frac{1}{2} \log(\sigma^2/D)$ to 10^{-16} (gate S1.1); on Figure 6(b) the cells at distortion D are the line $s = \sqrt{D}$ and the rate is the mean level along it.*

The hyperbolic geometry of the location–scale leaves (G6) is stated in the supplement (Appendix S.C); Figure 6(b) draws the radial geodesics and the calibrated circles under the prior. The Fisher-weighted transport geometry $\ell_\star^2 \mathcal{T}_{h,m} g_{W_{2,h}}$ is a distinct object from the Wasserstein–Fisher–Rao metric of unbalanced transport, where the Fisher term enters the transport itself. It is

also a classical construction read on a known action: the functional $\int[\frac{1}{2}|\dot{\mu}|^2 + c\mathcal{I}_{h,m}]$ is the action of the Schrödinger problem (Léonard, 2014), and the calibrated length is its Jacobi–Maupertuis length at zero energy (Arnold, 1989), with $\ell_\star^2 = 2c$ (Appendix S.B.2).

5 Resolution as an ordered family of channels

A finite description is a channel, and resolutions are ordered by post-processing. Figure 7 shows the four ledgers on a Blackwell-ordered tower of Gaussian sensors of the gyrator’s state; the statements follow.

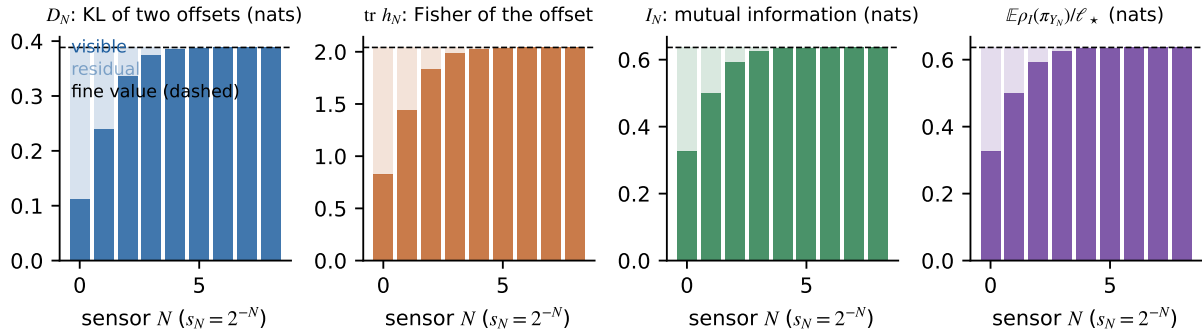


Figure 7: Resolution ledger on a Gaussian sensor tower. Blackwell-ordered Gaussian sensors $Y_N = x + \mathcal{N}(0, 2^{-2N}I)$ on the gyrator’s state with an unknown force offset: the relative information between two offsets, the Fisher information of the offset, the mutual information and the mean calibrated radius of the posteriors fill monotonically with N (dark: visible, light: residual) towards the fine value (line).

Theorem R1 (T9, Resolution Ledger Theorem: the tower and its ledgers · kernel · PROVED). *A resolution K_N is coarser than K_M when*

$$K_N = T_{N \leftarrow M} \circ K_M \quad (5.1)$$

for a channel T free of λ , Blackwell’s garbling order (Blackwell, 1953), so that $\Lambda \rightarrow Y_M \rightarrow Y_N$ is a Markov chain; a deterministic partition, where the coarse reading is a function of the fine one, is the special case. Along the order, $D_N \leq D_M$, $h_N \preceq h_M$ and $I_N \leq I_M$, by Equation (3.2), Equation (4.2) and B1, and at every level the applicable readings of the primitive split into a visible and a residual part, each under the conditions of its line (Table 3, R2). Two resolutions unrelated by post-processing are unordered: a sensor on one coordinate at noise 0.3 and a sensor on both at noise 0.6 dominate each other in different directions (gate X2.3); in time, the entropy production seen by an observer of the gyrator’s two coordinates grows monotonically along the dyadic steps $\Delta = 2^{-n}$ (gate LIB.7).

Proof. With $K_N = T \circ K_M$, the data-processing inequality of Equation (3.2) gives $D_N \leq D_M$, the score projection of Equation (4.2) gives $h_N \preceq h_M$, and B1 gives $I_N \leq I_M$; the four splits are Equation (3.2), Equation (4.2), the chain rule of mutual information and, under the radial identity, $\mathbb{E} \rho_{I,N} = \ell_\star I_N$ (Theorem G3); they are collected in R2. $\square$

Step by step, the increments of the ledger are the conditional ledgers: $D_{n+1} - D_n = \mathbb{E} D(P_{Y_{n+1}|Y_n} \| R_{Y_{n+1}|Y_n})$, $h_{n+1} - h_n = \mathbb{E}[(S_{n+1} - S_n)^{\otimes 2}]$, $I_{n+1} - I_n = I(\Lambda; Y_{n+1} | Y_n)$ and $\mathbb{E} \rho_{I,n+1} - \mathbb{E} \rho_{I,n} = \ell_\star I(\Lambda; Y_{n+1} | Y_n)$, by the chain rules of Equation (3.2), Equation (4.2) and B1: an ordered tower carries an additive spectrum of information revealed from coarse to fine (R2; on the sensors the increments equal the conditional quantities to $3 \cdot 10^{-12}$ in absolute value, gate X2.8b).

level	visible	residual	needs
relative information	D_N	$D_N^{\text{res}} = D - D_N$	any channel (K2)
Fisher	h_N	$h_N^{\text{res}} = \mathbb{E} \text{Var}(S Y_N)$	a regular stratum, scores in L^2 (O3)
Bayesian	$I_N = I(\Lambda; Y_N)$	$I_N^{\text{res}} = I - I_N$	a latent law under the prior (B1)
calibrated radius	$\mathbb{E}\rho_{I,N} = \ell_\star I_N$	$\ell_\star I_N^{\text{res}}$	the prior as reference and the radial identity (G3)

Table 3: The Resolution Ledger: the four lines of a tower and what each needs (R2; on the ordered sensors of Figure 7 the four quantities increase strictly and $\mathbb{E}\rho_I = \ell_\star I_N$ to 10^{-16} , gate X2.1).

Theorem R5 (T10, Resolution Recovery Theorem · kernel · PROVED). *Let $\mathcal{F}_1 \subset \mathcal{F}_2 \subset \dots$ be a filtration generating $\mathcal{F}$ up to $(P + R)$ -null sets, R a probability. Then $D(P|_{\mathcal{F}_N} \| R|_{\mathcal{F}_N}) \uparrow D(P \| R)$ in $[0, \infty]$ and the residual term decreases to zero. For a family differentiable in quadratic mean with scores in L^2 , $h_N \uparrow h$ in the Loewner order with $h_M - h_N = \text{Cov}(S_M - S_N)$. For observations Y_N generating Y , $I(\Lambda; Y_N) \uparrow I(\Lambda; Y)$, and under the radial identity $\mathbb{E}\rho_{I,N} \uparrow \mathbb{E}\rho_I$. The proof, in Appendix S.B, rests on Lévy’s upward theorem and on relative entropy as a supremum over finite partitions. On the gyrator the dyadic tower in time is a filtration on path space: the observer of both coordinates sees 99.4% of σ at $\Delta = 2^{-8}$ (gate X5.2); the tower of sensors is a garbling with $K_N P \rightarrow P$ and $K_N R \rightarrow R$, where data processing bounds the relative-information line above and lower semicontinuity bounds it below, while the Fisher, Bayesian and radial lines are read on the closed forms of the Gaussian realization: the four ledgers are within 0.5% of their fine values from $N = 5$, with a gap linear in the noise variance (gate X2.4).*

Theorem G10 (T11, Almost-Sufficient Stability Theorem · kernel · PROVED). *Let $\delta^2 h \preceq h_K \preceq h$ on the same stratum, $0 < \delta \leq 1$, with the same reference measure m and the same gauge $\ell_\star$, the coarse metric h_K being that of a δ -almost sufficient statistic (Yamaguchi and Nozawa, 2024). Then $\delta W_{2,h} \leq W_{2,h_K} \leq W_{2,h}$, $\mathcal{I}_{h,m} \leq \mathcal{I}_{h_K,m} \leq \delta^{-2} \mathcal{I}_{h,m}$, and*

$$\delta L_I^h(\gamma) \leq L_I^{h_K}(\gamma) \leq \delta^{-1} L_I^h(\gamma), \quad \delta d_I^h \leq d_I^{h_K} \leq \delta^{-1} d_I^h : \quad (5.2)$$

almost sufficient in Fisher means almost isometric in the slope geometry, and $\delta^2 = 1 - \varphi$ with φ the largest fraction of missing information of the projection. Under HR0 for (Λ, h_K, m) as well, $L_I^{h_K}$ is the calibrated length of the coarse experiment and the radial identity holds with the same right-hand side. On the gyrator the 32-step sub-window of the 64-step window gives $\delta^2 = 0.49$ at the nominal parameter, a pointwise reading of the sandwich, while one coordinate alone gives $\delta = 0$ (gate X1.5); on the tower of sensors $\delta_N^2 = \min_i \lambda_i(C)/(\lambda_i(C) + 2^{-2N})$ in closed form and the calibrated length of a translation of a Gaussian belief stays within $[\delta_N, \delta_N^{-1}]$ of its fine value and tends to it (gate X2.5). Proof in Appendix S.B.

Corollary G11 (Uniform almost-sufficiency convergence and pipeline composition of reduction bounds · kernel · PROVED). *Under the hypotheses of Theorem G10: (i) if a tower (h_N) on the same stratum satisfies $\delta_N^2 h \preceq h_N \preceq h$ with $\delta_N \rightarrow 1$, then $L_I^{h_N}(\gamma) \rightarrow L_I^h(\gamma)$ for every common admissible curve of finite length and $d_I^{h_N}(\mu, \nu) \rightarrow d_I^h(\mu, \nu)$ for every pair at finite distance: non-radial convergence is automatic along uniformly almost-sufficient towers, and what remains open is the convergence of minimizing geodesics and the discrete realizations that leave the stratum (Q2); (ii) if K_1 is δ_1 -almost sufficient and K_2 is δ_2 -almost sufficient for the model induced by K_1 , then $K_2 \circ K_1$ is $\delta_1 \delta_2$ -almost sufficient, $(\delta_1 \delta_2)^2 h \preceq h_{K_2 K_1} \preceq h$, so that $-\log \delta$ is an additive upper budget of geometric distortion along a reduction pipeline and the fractions of missing information compose as $\varphi_{\text{tot}} \leq 1 - \prod_i (1 - \varphi_i)$. Both follow from Equation (5.2), by passing to the limit and by chaining the Loewner sandwiches. On the tower of sensors $|L_N/L_\infty - 1| \leq \delta_N^{-1} - 1 \rightarrow 0$, and δ^2 composes multiplicatively as a lower bound (gates X2.6, X2.7).*

Proof. (i) Equation (5.2) with $\delta = \delta_N$ gives $\delta_N L_1^h \leq L_1^{h_N} \leq \delta_N^{-1} L_1^h$ and the same sandwich for d_1 ; let $\delta_N \rightarrow 1$. (ii) $h_{K_2 K_1} \succeq \delta_2^2 h_{K_1} \succeq \delta_2^2 \delta_1^2 h$ by the two Loewner sandwiches, and $1 - \varphi_{\text{tot}} \geq \prod_i (1 - \varphi_i)$ is the same inequality read with $\delta_i^2 = 1 - \varphi_i$. $\square$

Theorem K5 (T13, Irreversibility Ledger Theorem · kernel · PROVED). *When a reversal $P^\dagger$ equivalent to P exists, $\Sigma = \log dP/dP^\dagger$ satisfies $\mathbb{E} e^{-\Sigma} = 1$ and $\mathbb{E} \Sigma = D(P \| P^\dagger) \geq 0$; under every quotient K of the paths that commutes with time reversal, $K(P^\dagger) = (KP)^\dagger$, Equation (3.2) applied to the pair $(P, P^\dagger)$ splits the arrow exactly,*

$$\langle \Sigma_{\text{fine}} \rangle = \langle \Sigma_{\text{visible}} \rangle + \mathbb{E}_{KP} D(P_{X|Y} \| P_{X|Y}^\dagger), \quad \langle \Sigma_{\text{visible}} \rangle = D(KP \| (KP)^\dagger), \quad (5.3)$$

so that per unit time $\sigma = \sigma_{\text{visible}} + \sigma_{\text{res}}$ with $\sigma_{\text{res}} \geq 0$, and the visible irreversibility obeys $\langle \Sigma_{\text{visible}} \rangle \leq \langle \Sigma_{\text{fine}} \rangle$ (Kawai et al., 2007; Gomez-Marin et al., 2008). On the gyrator every scalar linear observable is a stationary Gaussian process, hence reversible (Weiss, 1975), the window covariance being symmetric Toeplitz: an observer of one coordinate reads zero, and the two coordinates read $\sigma = 0.2793$, measured as 0.2839 ± 0.0037 on $2 \cdot 10^4$ paths (gates X5.1, X5.3, X5.4, LIB.6, SYM.5). Residual irreversibility under coarse graining is a subject of its own (Degünther et al., 2024).

Proof. $\mathbb{E}_P e^{-\Sigma} = \int \frac{dP^\dagger}{dP} dP = 1$ and $\mathbb{E}_P \Sigma = D(P \| P^\dagger) \geq 0$. When K commutes with time reversal, $D(KP \| (KP)^\dagger) = D(KP \| KP^\dagger) \leq D(P \| P^\dagger)$ by Equation (3.2) applied to the pair $(P, P^\dagger)$. $\square$

Corollary R6 (Irreversibility Recovery · kernel · PROVED). *A corollary of Theorem K5 and Theorem R5. Let $K_1, K_2, \dots$ be quotients of path space forming a refinement order, each commuting with time reversal, $K_N(P^\dagger) = (K_N P)^\dagger$. The visible irreversibility rates $\sigma_N = D(K_N P \| (K_N P)^\dagger)/t$ grow along the order and stay at or below the fine rate σ ; when K_N records exactly the σ -algebra $\mathcal{F}_N$ of an increasing filtration generating the path σ -algebra up to $(P + P^\dagger)$ -null sets, $\sigma_N \uparrow \sigma$.*

Proof. Monotonicity along the order is Theorem K5 applied to the pair of quotients. For the filtration, apply Theorem R5 with the reversal $P^\dagger$ as reference, which is a probability: $D(P|_{\mathcal{F}_N} \| P^\dagger|_{\mathcal{F}_N}) \uparrow D(P \| P^\dagger)$, and divide by t . $\square$

Check. On the dyadic tower in time of the gyrator the visible production per unit time increases as the observation step decreases and reads 0.2777 at $\Delta = 2^{-8}$ against $\sigma = 0.2793$, that is 99.4% (gate X5.2).

Scope of the constraint. The constraint carried by thermodynamic consistency is the admissibility of the coarse description: its quotient commutes with time reversal and its declared ports balance. The two inequalities then follow, the visible production of a level staying under the fine value and above that of any coarser level, and Theorem G10 bounds the distortion that the projection introduces, which is what turns a coarse description into a controlled object.

Harmonic Markov resolution. A resolution tower is any ordered family of channels. Under HR0 the heat semigroup of the reference, $K_t = e^{tL_m}$ with $L_m = \Delta_h + \nabla_h \log w \cdot \nabla_h$, is one: each K_t is a Markov channel, $K_{t_2} = K_{t_2-t_1} \circ K_{t_1}$ for $t_2 \geq t_1$, so that increasing t is a Blackwell coarsening, and along $\mu_t = K_t \mu$, $-\frac{d}{dt} D(\mu_t \| m) = \mathcal{I}_{h,m}(\mu_t)$ (G1): relative Fisher information is the instantaneous relative-information loss rate of the canonical harmonic coarse-graining. When the generator has a compact resolvent the semigroup is diagonal in its eigenbasis, $K_t \phi_k = e^{-\lambda_k t} \phi_k$, coarse to fine as $\lambda_1 \leq \lambda_2 \leq \dots$; the finite spectral cutoff is the analysis, with the χ^2 ledger of Parseval and a tail bounded by $e^{-2\lambda_{N+1} t}$, and the exponential family of the first N modes is the positive reconstruction (R3, R4, in Appendix S.C; diffusion maps build multiscale coordinates from the same operators, Coifman and Lafon, 2006). The kernel is free of this choice.

6 Emergent finite structure

Here, and only here, appear meshes, cells, facets, graphs, hypergraphs, symbols and Markov factorizations: they are what a resolution reads on a geometry, and a graph used as a finite representation of a continuum geometry is determined by the measure, the declared resolution and the discretization rule as much as by the geometry.

Cells and facets. Quantised on cells of step ε , the current of a process crosses facets and the balance of each cell is a discrete Kirchhoff law; the chain with facet rates k_{ij} and stationary law p has the entropy production

$$\sigma_\varepsilon = \frac{1}{2} \sum_{ij} J_{ij} A_{ij}, \quad J_{ij} = p_i k_{ij} - p_j k_{ji}, \quad A_{ij} = \log \frac{p_i k_{ij}}{p_j k_{ji}}, \quad (6.1)$$

which approaches the production of the process as $\varepsilon \rightarrow 0$ on the gyrator with facet rates of Scharfetter–Gummel type, the general convergence of such gradient-flow structures being the subject of Maas (2011) and Forkert et al. (2022); under the tested eight-neighbour mobility convention, the diagonal links add a persistent excess production, so the ledger is a property of the facet conductances of the stencil (E2, Figure 8).

Three graphs and the curvature of the ruler. On one object live three graphs: the proximity graph of the cells, the interaction hyperedges of the generator and the conditional-independence graph of the law; on the gyrator the first ignores the coupling k while the two others depend on it (E3). The law ruler joins them: on the stratum $(\log k_1, \log k_2)$ at fixed k , the stationary law factorizes at $k = 0$ with one parameter per factor, so the Fisher metric is the direct sum of two one-dimensional metrics, $\frac{1}{2}I$ in log-stiffness coordinates and flat; it curves for $k > 0$, with Gaussian curvature -0.274 at the nominal point for $k = 0.6$; the generator edge A_{12} , the conditional-independence edge $(C^{-1})_{12}$ and the curvature of the ruler leave zero together with the coupling, three concomitant signatures of the same coupling in this Gaussian realization, the same loss of factorization that the coupling term of Equation (6.2) measures (gates SYM.7, X6.6; Figure 9). A graphical factorization is a realization property (Lauritzen, 1996).

Two axes of information. Relative to a product reference, what a belief has learned splits into what concentrates each coordinate and the total correlation that couples them,

$$D(\mu \parallel m) = \sum_i D(\mu_i \parallel m_i) + D(\mu \parallel \prod_i \mu_i), \quad (6.2)$$

and the two terms move separately in the Gaussian tests of E4: on a factorized prior a coordinate-wise sensor moves the marginal part while the coupling part stays at zero, a joint sensor moves the coupling part, and on a coupled prior a precise reading of one coordinate drives the coupling part down.

Theorem E6 (T12, Predictive Closure Ledger Theorem · kernel · PROVED). *Let (x_t) be stationary, q measurable with values in a finite alphabet, $X_t = q(x_t)$, and at the time resolution Δ let $I_{\text{pred}} = I(X_{t+\Delta}; x_t)$, $I_{\text{ret}} = I(X_{t+\Delta}; X_t)$ and $I_{\text{res}} = I(X_{t+\Delta}; x_t \mid X_t)$. Then*

- (i) $I_{\text{pred}} = I_{\text{ret}} + I_{\text{res}}$, with $I_{\text{res}} = \mathbb{E}_{x_t} D(P_{X_{t+\Delta}|x_t} \parallel P_{X_{t+\Delta}|X_t})$;
- (ii) $I_{\text{res}} = 0$ exactly when $X_{t+\Delta}$ and x_t are conditionally independent given X_t : X_t is a predictively sufficient statistic of x_t for $X_{t+\Delta}$;
- (iii) $0 \leq I_{\text{res}} \leq I_{\text{pred}} \leq H(X_t)$; wherever $I_{\text{pred}} > 0$ and $H(X_t) > 0$, the loss $\varepsilon_{\text{cl}} = I_{\text{res}}/I_{\text{pred}}$, the fidelity $\eta_{\text{ret}} = I_{\text{ret}}/I_{\text{pred}} = 1 - \varepsilon_{\text{cl}}$ and the memory $\eta_{\text{mem}} = I_{\text{pred}}/H(X_t)$ lie in $[0, 1]$;
- (iv) $\mathbb{E} \|P_{X_{t+\Delta}|x_t} - P_{X_{t+\Delta}|X_t}\|_{\text{TV}} \leq \sqrt{I_{\text{res}}/2}$;
- (v) when the paths are continuous and x_t avoids the boundaries of q almost surely, $I_{\text{ret}}, I_{\text{pred}} \rightarrow H(X_t)$ and $I_{\text{res}} \rightarrow 0$ as $\Delta \rightarrow 0$.

A level closes at tolerance (η, α) when the three ratios are defined, $\varepsilon_{\text{cl}} \leq \eta$ and $\eta_{\text{mem}} \geq \alpha$. On the double well $U(x) = E_b(x^2 - 1)^2$ with the quotient {left, right}: at $T = 0.15$ the loss stays at or below

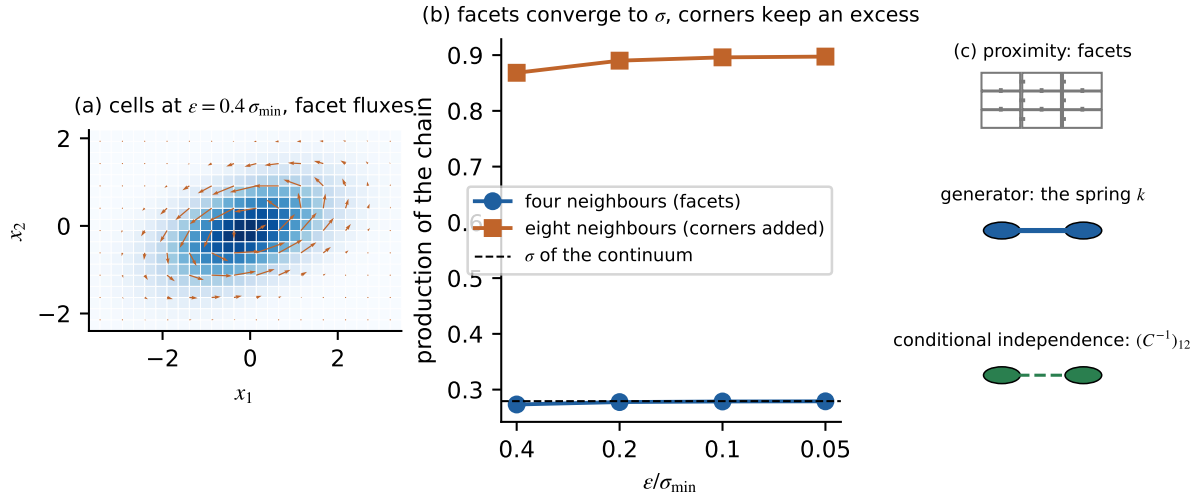


Figure 8: Cells, facets and the finite-volume ledger. (a) The stationary gyrtor quantised at $\varepsilon = 0.4\sigma_{\min}$, the current through the facets. (b) The production of the four-neighbour chain converges to σ ; the eight-neighbour chain keeps an excess. (c) Three graphs on one object: the proximity graph of the cells, the interaction hyperedge of the spring, the conditional-independence graph of the stationary law.

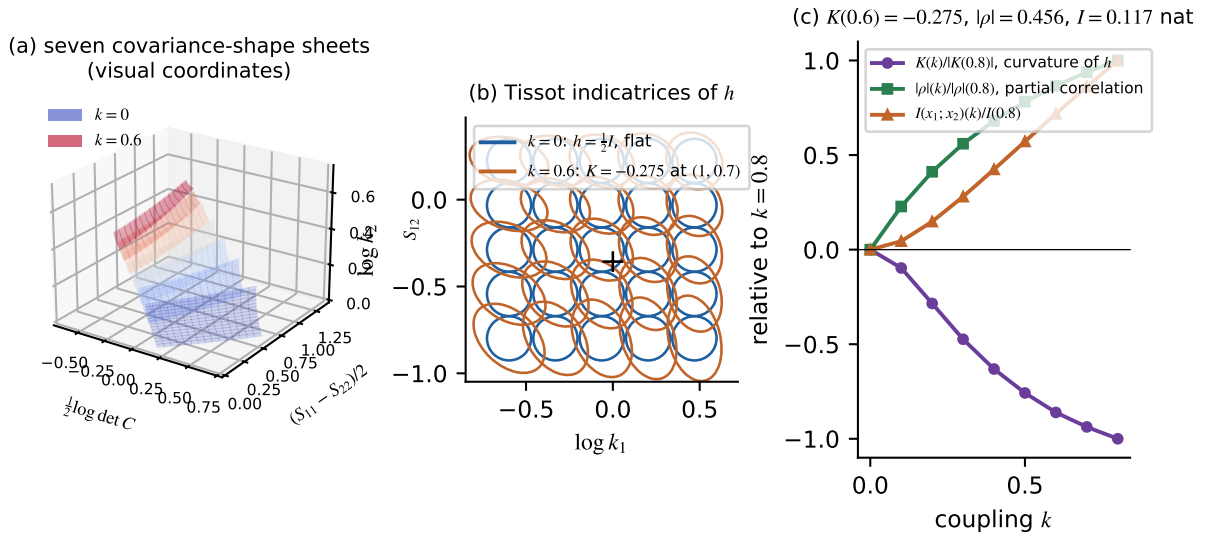


Figure 9: Coupling, curvature and the three graphs. (a) Seven covariance-shape sheets of the stationary gyrtor, $k = 0$ (blue) to $k = 0.6$ (red), each the image of the chart $(\log k_1, \log k_2)$ in the coordinates $(\frac{1}{2} \log \det C, (S_{11} - S_{22})/2, S_{12})$, $S = C/\sqrt{\det C}$: visual covariance-shape coordinates, free of any isometry claim; the curvature is computed intrinsically from the Fisher metric. (b) Tissot indicatrices of the law ruler h on the chart, $T_1 = 2$, $T_2 = 0.5$: at $k = 0$ the metric is $\frac{1}{2}I$, circles everywhere, curvature zero; at $k = 0.6$ the indicatrices stretch and turn, curvature -0.274 at the cross. (c) Against the coupling k : the curvature of h , the partial correlation and the mutual information of the two coordinates, each relative to its value at $k = 0.8$, leave zero together. The curvature is a local quantity, -0.35 and -0.20 at two other points of the chart.

0.005 and the memory above 0.79 on $\Delta \in [0.5, 20]$; at $T = 1$ the fidelity stays near 0.88 while the memory falls to 0.04 at $\Delta = 2$ and to the floor of the estimator beyond; the mean total-variation error of predicting from the symbol is 0.001 at $T = 0.15$ and 0.075 at $T = 1$, $\Delta = 0.5$; the short-lag limit is the occupation entropy (gates X8.5, X8.6b, X8.7, X8.8, X8.2b; Figure 10). Proof in Appendix S.B; the retained information is the relevant information of the information bottleneck (Tishby et al., 1999), $I_{\text{res}} = 0$ is the informational closure of Bertschinger et al. (2006) and Pfante et al. (2014), and predictive sufficiency is the causal-state construction of Shalizi and Crutchfield (2001); see Geiger (2026) for the reduction of Markov chains.

Realization E5 (Symbol $\cdot$ realization $\cdot$ MEASURED). The number of live cells of a belief at resolution ε is its perplexity e^{H_ε} , equal to $2\pi e \sqrt{\det C} / \varepsilon^2$ for a Gaussian to 0.1 % (gate X6.4b), and informative experiments reduce its logarithm, the entropy H_ε , on average. A symbol at resolution $(\varepsilon, \Delta, \eta, \alpha)$ is a coarse state that is predictively closed to tolerance (η, α) over the horizon Δ . With atoms in the reference, exact certainty has a finite relative-information cost, $-\log m(\{x\})$, where the continuous horizon has an infinite one; on one well of the double well the count of cells above 10^{-3} of the largest cell mass shrinks from 16 to 7 as T goes from 0.4 to 0.05, the perplexity of the cell masses from 11.1 to 3.5 (gates X8.3, PUTS.1), and the exit rate follows Arrhenius (E7).

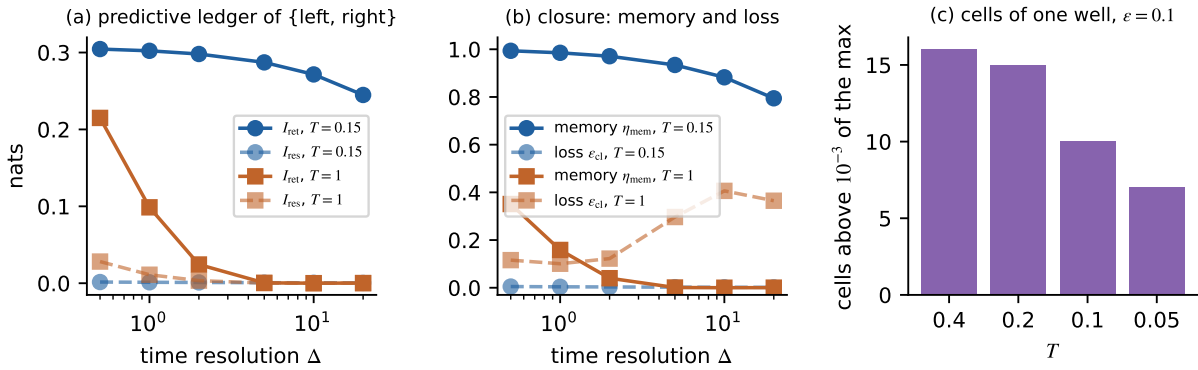


Figure 10: Double well: predictive closure and symbols. (a) The predictive ledger of the two-symbol quotient against the time resolution, retained and residual information at $T = 0.15$ (closes) and $T = 1$ (drowned barrier). (b) The closure ratios, memory η_{mem} and loss ε_{cl} : at $T = 1$ the loss stays small while the memory dies. (c) The count of cells of one well above 10^{-3} of the largest cell mass at fixed spatial resolution, against the temperature.

7 Physical realization and numerical experiments

Scope of the three benchmarks. The benchmarks of this section differ in nature and share a bookkeeping. The Brownian gyrator is a coupled diffusion between two thermal baths whose stationary current circulates and whose scalar marginals are reversible. The driven ring carries a harmonic current sustained by a work port, and its zone balance has two terms that stay away from zero. The metastable double well carries symbols and closes predictively. Each is read with the same channel ledger, the same irreversibility ledger and the same port balance, which is what thermodynamic consistency asks of a framework (Section 2).

7.1 Informational dissipative structures

Definition H1 (Informational dissipative structure $\cdot$ realization $\cdot$ DEFINITION). A realization observed at a declared resolution (ε, Δ) is an informational dissipative structure when (i) its quotient closes at tolerance (η, α) , $\varepsilon_{\text{cl}} \leq \eta$ and $\eta_{\text{mem}} \geq \alpha$, over a declared interval of horizons: under the hypotheses of E6 (v), every finite quotient with $H(X_t) > 0$ has $\varepsilon_{\text{cl}} \rightarrow 0$ and $\eta_{\text{mem}} \rightarrow 1$

as $\Delta \rightarrow 0$, so a single short lag carries zero content; (ii) the path law of the level is irreversible, $D(P \parallel P^\dagger) > 0$, which, for the stationary overdamped diffusions used here under the standard time reversal, is equivalent to a nonvanishing stationary current (K5); (iii) the irreversibility is sustained by declared environmental ports whose conservation ledger and entropy-flow ledger close: in the stationary state $\dot{W} = \sum_i \dot{Q}_i$ and $\sigma = \sum_i \dot{Q}_i/T_i$, with $\dot{W}$ the work received from the work ports, $\dot{Q}_i$ the heat delivered to bath i , and $\dot{W} = 0$ for a purely thermal realization (H2). Its revision is priced by $\sqrt{\mathcal{I}_{h,m}}$ (G2). The gyrator satisfies clause (ii) exactly when $k(T_1 - T_2) \neq 0$, its port balances of clause (iii) closing for the realization and carrying a flux under the same condition, and everything is a closed form: with $\Gamma = \mathbf{D}C^{-1} - A$ the current is $j = \Gamma x p$, $\Gamma C = sJ$ with

$$s = \frac{k(T_1 - T_2)}{2k + k_1 + k_2}, \quad \sigma = \frac{k^2(T_1 - T_2)^2}{T_1 T_2 (2k + k_1 + k_2)}, \quad Q_1 = -Q_2 = -\frac{k^2(T_1 - T_2)}{2k + k_1 + k_2}, \quad (7.1)$$

the current is the rotation of the stream function $\psi = sp$, a sufficient and exactly solvable realization of clause (ii), and the three quantities vanish exactly when $k(T_1 - T_2) = 0$ (gates X4.2, X4.3, X4.4, LIB.1, LIB.2, LIB.4, LIB.5, SYM.1–SYM.4, SYM.6). The driven ring carries a harmonic current, $\sigma = F^2/T$ at uniform potential, so clause (ii) covers coexact and harmonic currents alike; its drive is a work port, $\dot{W} = FjL$, delivered as heat to its single bath, $\sigma = \dot{W}/T$ (gate RING.3).

Realization H2 (Ports · realization · MEASURED). A thermodynamic port is a declared environmental channel through which a realization exchanges a conserved quantity or a control signal: a thermal port, a work or control port. An observation channel belongs to the operational access, and becomes a port once its physical implementation joins the realization. In the canonical thermal realization a bath at temperature T selects a reference proportional to $e^{-E/k_B T}$ against the declared base measure, with $k_B = 1$ in the units of the gyrator, and for an isothermal port under local detailed balance the heat delivered is $k_B T$ times the nats given to the port; a work port carries $\dot{W}$ with zero entropy flow, and the stationary energy ledger reads $\dot{W} = \sum_i \dot{Q}_i$, with $\dot{W}$ received from the work ports and $\dot{Q}_i$ delivered to bath i . On the gyrator the heat rates delivered to the baths satisfy $\dot{Q}_1 + \dot{Q}_2 = 0$ and $\sigma = \dot{Q}_1/T_1 + \dot{Q}_2/T_2$, Clausius, exactly (gate LIB.3); a geometric boundary is one way to realize a port. The reference potential $-\log(dm/d\nu)$ of the kernel becomes $\beta E + \text{const}$ in this realization alone (K6).

Realization E1 (Two-Ledger Identity: the balance of a zone · realization · PROVED). On a weighted manifold with a density f relative to m and a current $j = fv$, every zone A obeys

$$\frac{d}{dt} \int_A f \log f \, dm = -\Phi_A^{\text{rel}} + B_A^{\text{rel}}, \quad \Phi_A^{\text{rel}} = \oint_{\partial A} (1 + \log f) \langle j, n \rangle d\sigma_m, \quad B_A^{\text{rel}} = \int_A \langle v, \nabla \log f \rangle d\mu, \quad (7.2)$$

content, interface flux and bulk relative-information term; for the gradient flow $v = -\nabla \log f$ the bulk term is $-\mathcal{I}_A$. For a diffusion of drift b and constant diffusion matrix $\mathbf{D}$, the thermodynamic entropy production of the zone is

$$\sigma_A = \int_A \langle v, \mathbf{D}^{-1} v \rangle d\mu = M_A - B_{m,A}^{\text{rel}} - \int_A \langle v, \nabla \log m \rangle d\mu, \quad M_A = \int_A \langle v, \mathbf{D}^{-1} b \rangle d\mu, \quad (7.3)$$

with M_A the entropy flow to the medium in the overdamped thermodynamic realization of H2, and a reference term that vanishes for the volume reference, since $v = b - \mathbf{D} \nabla \log p$ and $\nabla \log p = \nabla \log f + \nabla \log m$ (Appendix S.B.1): the relative-information balance and the thermodynamic balance are two ledgers, and the bulk term depends on the declared reference. On the stationary gyrator Φ_A^{rel} and B_A^{rel} vanish on every zone, the current running along the level sets of the density, while $\sigma_A = 0.172$ on the disk of Figure 11 equals the medium term (gates X4.1b, X4.5); on the driven ring $dx = (F - U'(x))dt + \sqrt{2T}dW$ with $U = \cos x$, $F = 1$, $T = 0.5$, the zone $[0, \pi]$ has $\Phi_A^{\text{rel}} = B_A^{\text{rel}} = 0.109$ and $\sigma_A = 0.916$ equal to the medium term minus the bulk term (gates RING.1, RING.2; with the Gibbs reference the bulk term changes and the identity holds with its reference term, gate RING.4; closed forms in Appendix S.A, Figure 13).

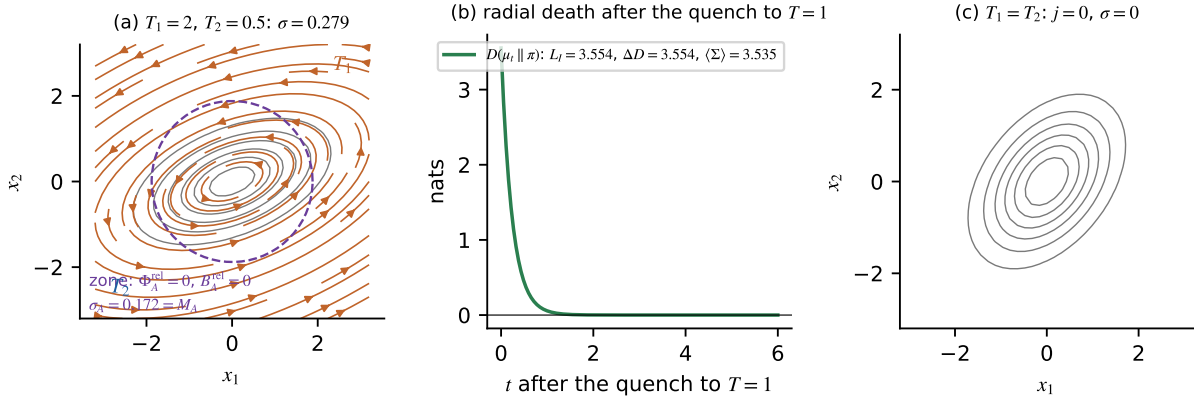


Figure 11: Gyrator: current, ports and the two ledgers. (a) At $T_1 = 2, T_2 = 0.5$: $\sigma = 0.279$ the stationary current (orange) runs along the level sets of the density (grey); on the disk (dashed) the interface flux and the bulk term of the relative-information balance vanish while the entropy production is positive, $\Phi_A^{\text{rel}} = 0, B_A^{\text{rel}} = 0, \sigma_A = 0.172$; the two baths are the ports. (b) At equal temperatures $T_1 = T_2 = 1$, from the displaced stationary law $\mathcal{N}(m_0, C)$ of Theorem G3, the ensemble relaxes along the radial geodesic: at the gauge of the experiment, $\ell_\star = 1$, the calibrated length of its path, the drop of relative information and the measured entropy production coincide. (c) At equal temperatures the current and the production vanish.

7.2 Numerical experiments

Protocol. Every gate was declared in writing before its first computation. Everything runs on ten CPU cores. The gates are closed forms or exact propagations, with two sampled exceptions: the entropy productions of K5 and G3, averaged over $2 \cdot 10^4$ Euler–Maruyama paths, and the double-well statistics of E5, E6 and E7, time averages along single Euler–Maruyama trajectories with fixed seeds, quoted without error bars. Appendix S.D names every gate, grouped by family.

What the benchmarks test. The residual arrow: every scalar linear observable is reversible, the window covariance being symmetric Toeplitz, while the two coordinates read $\sigma = 0.2793$, measured as 0.2839 ± 0.0037 on $2 \cdot 10^4$ paths (K5, Figure 12). Fisher identifiability: rank four from one coordinate, five from two (Figure 5). The stationary current and the two ledgers of a zone (Realization E1, Figure 11). The finite-volume projection (E2). The relaxation at equal temperatures as a radial geodesic (G3). The double well tests the predictive closure and the symbols (E6, E5, E7). The gyrator is deliberately minimal and therefore degenerate in useful ways: its stationary bulk term vanishes pointwise and every scalar linear marginal is reversible; these are stress tests of observability, and the driven ring supplies the missing nondegenerate balance.

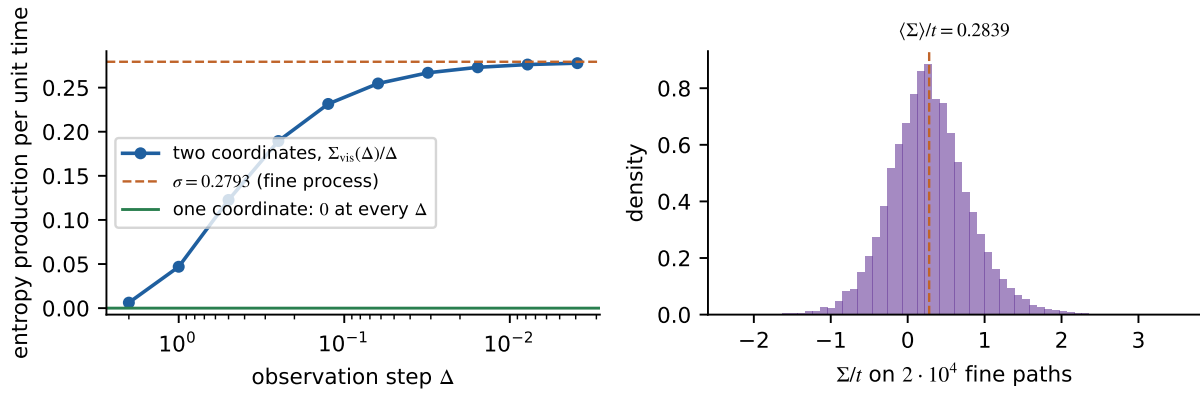


Figure 12: Visible irreversibility against time resolution. Left: the entropy production seen by the observer of both coordinates grows with its time resolution and tends to σ , an instance of exhaustive recovery on path space; the observer of one coordinate reads zero at every Δ . Right: the entropy production of the fine process on $2 \cdot 10^4$ paths.

8 Discussion

8.1 Relation to prior work

Table 4 says, for each named theorem, what is classical, what is read anew and what is proved here.

result	status	provenance
T1 Declared-Reference Principle	principle	this work (the datum and its references)
T2 Channel Ledger Theorem, C1	imported	Kullback, Csiszár; Leinster, Baez–Fritz for the characterization
T3 Retention Theorem	imported, proof recalled	Csiszár, Zellner; the one-line Gibbs identity
T4 Recovery-Kernel Theorem	imported	Blackwell
T5 Operational Fisher Theorem	imported	Čencov, Ay–Jost–Lê–Schwachhöfer
T6 Missing-Fisher Ledger Theorem	classical identity, IDS reading	Fisher, Orchard–Woodbury, Louis, Meng–Rubin
T7 Slope-Geometry Proposition	proved here, two lines	strong upper gradients and curves of maximal slope of Ambrosio–Gigli–Savaré
T8 RESG Theorem	proved here on a classical action	the calibrated length is the Jacobi–Maupertuis length of the Schrödinger action at zero energy (Arnold; Léonard); extends Caraffa (2026) (Def. 3.6, Thms 4.10, 4.15) from the entropy of a density to relative entropy under a declared reference, with the radial identity; the Wasserstein calculus of entropy imported
T9 Resolution Ledger Theorem	assembly proposed here	the four chain rules are classical, their common ledger is this work
T10 Resolution Recovery Theorem	classical for D and I , proved here for Fisher and radius	Pinsker’s supremum, Barron; Lévy’s upward theorem
T11 Almost-Sufficient Stability Theorem	proved here	a lift of the bi-Lipschitz equivalence of Yamaguchi–Nozawa to $W_{2,h}$, $\mathcal{I}_{h,m}$ and d_I
T13 Irreversibility Ledger Theorem, R6	derived here, a data-processing consequence	path-space relative entropy and data processing; the coarse-grained irreversibility of Kawai et al., Gomez-Marin et al., Degünther et al.
T12 Predictive Closure Ledger Theorem	classical chain rule	informational closure of Bertschinger et al. and Pfante et al.; Shalizi–Crutchfield; assembled here: the tolerance pair (η, α) and a Pinsker control of the prediction error
Hyperbolic-Leaf Theorem (G6), G11	G6 restated, proof recalled; G11 proved here	Caraffa (2026, Thms 4.18–4.19); flat W_2 leaf, J_0/s^2 scaling, Stam; Loewner sandwiches
realizations, graphs, cells, symbols	measured	this work, gyrator, ring and double well

Table 4: Relation to prior work: status and provenance of each named theorem.

8.2 Scope and limitations

Table 5 says what each statement needs.

statement	needs
exact channel ledger (K2)	a measurable channel
Fisher geometry (O1, O3)	a regular identifiable stratum, scores in L^2 , an observation channel free of the parameter
slope geometry (G3)	the contract HR0
radial identity (G5)	the prior as reference and an entropy-ergodic flow
exhaustive recovery (R5)	a generating filtration; scores in L^2 for Fisher
stability (G10)	a δ -almost sufficient projection, the same reference measure
predictive closure (E6)	a stationary process and a finite alphabet; path continuity and boundary avoidance for the short-lag limit
harmonic resolution (R3)	a compact resolvent
irreversibility ledger (K5)	an equivalent time-reversed path law and a quotient commuting with time reversal
irreversibility recovery (R6)	a generating refinement with the same time-reversal compatibility
informational dissipative structure (H1)	predictive closure, path irreversibility and declared physical ports with closed balances
thermodynamic heat reading (H2)	a physical realization under local detailed balance

Table 5: What each statement needs.

The benchmarks leave untested, by construction, generic non-Gaussian beliefs, non-radial geodesics of the slope geometry and nonlinear observation channels. The contract HR0 is read on flat location–scale and Gaussian-location families and on the physical ensemble space; a curved identifiable quotient, such as the stratum of E3, carries zero calibrated-length measurement here. The clauses of H1 are tested on complementary benchmarks: the current and the ports on the gyrator and the driven ring, the predictive closure on the double well; an object that closes and carries a sustained current is the next benchmark. The open problems are stated in Appendix S.E.

9 Conclusion

The proposed contribution is a reference-explicit ledger calculus carried across three distinct spaces and through finite-resolution projections. Learning, diffusion models, scaling laws, Turing machines, the compiler, gauge theories, gravity and quantum channels are outside this paper.

IDS begins from a reduction, and a representation comes second. A declared channel determines which candidate laws are distinguishable; relative entropy records exactly what the channel carries and what stays residual. On regular identifiable families, differentiating that ledger yields the Fisher metric and its missing-information decomposition. Lifting the same operational metric to probability measures over identifiable laws turns the Wasserstein slope of relative entropy into a calibrated geometry of revision. Finite descriptions then form a tower of channels in place of a primitive graph: their validity is quantified by the information that survives the projection, and their closure by the predictive ledger. This separation changes the role of discrete structure. Cells, graphs, conditional-independence relations and physical ports are realization-level objects, outside the axioms of the informational kernel. The Brownian gyrator, the driven ring and the double well show how the same bookkeeping reads residual irreversibility, finite-volume projections, metastable symbols and predictive closure while keeping thermodynamic entropy production distinct from local relative-information balance. Thermodynamic consistency is the thread of the architecture: each physical realization carries its declared port balances, while the informational ledger and the irreversibility of time-reversal-compatible quotients are ordered by refinement and reach their fine values under the exhaustive hypotheses. The remaining mathematical frontier is precise: the convergence of non-radial calibrated lengths under discretization, the Hodge decomposition of residual irreversibility, and the extension of the operational geometry to stratified identifiable

quotients (Appendix S.E). These are questions about the fidelity of finite representations of a continuum geometry; the kernel assumes zero privileged graph.

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Supplement

S.A Closed forms

The gyrator. With $A = \begin{pmatrix} k_1+k & -k \\ -k & k_2+k \end{pmatrix}$ and $\mathbf{D} = \text{diag}(T_1, T_2)$, the stationary covariance solves $AC + CA = 2\mathbf{D}$; writing $\Delta = (2k + k_1 + k_2)(kk_1 + kk_2 + k_1k_2)$,

$$C_{11} = \frac{T_1k^2 + T_1kk_1 + 3T_1kk_2 + T_1k_1k_2 + T_1k_2^2 + T_2k^2}{\Delta}, \quad C_{12} = \frac{k(T_1k + T_1k_2 + T_2k + T_2k_1)}{\Delta},$$

and C_{22} by exchange of the two sides. $\Gamma = \mathbf{D}C^{-1} - A$ satisfies $\Gamma C = sJ$ with $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$; the stream function of the current is $\psi = sp$; the entropy production, the heats and Clausius are Equation (7.1) (gates SYM.1–SYM.4, symbolic, exact). The three quantities factor as $k(T_1 - T_2)$, $k^2(T_1 - T_2)^2$ and $k(T_1 - T_2)$ times a cofactor of constant sign on the positive orthant, so they vanish exactly when $k(T_1 - T_2) = 0$ (gate SYM.6). The window covariance of a scalar stationary process is symmetric Toeplitz, hence invariant under time reversal (gate SYM.5).

The driven ring. On the circle of length L , $dx = (F - U'(x))dt + \sqrt{2T}dW$ with $U(x) = a \cos(2\pi x/L)$ has the stationary density $p = g/Z$, $g(x) = e^{-\varphi(x)} \int_x^{x+L} e^{\varphi(y)} dy$, $\varphi = (U - Fx)/T$, and the constant current $j = T(1 - e^{-FL/T})/Z$ (Risken, 1989); the periodic extension gives $\int_x^{x+L} e^\varphi = (E(L) - E(x)) + e^{-FL/T}E(x)$ with $E(x) = \int_0^x e^\varphi$. For a zone $A = [a, b]$: $\Phi_A^{\text{rel}} = j \log(p(b)/p(a))$, $B_A^{\text{rel}} = \int_A j(\log p)' dx = \Phi_A^{\text{rel}}$, the content being stationary; $\sigma_A = \int_A j^2/(Tp) dx$ and the medium term $M_A = (j/T)[F(b-a) - U(b) + U(a)]$ satisfy $\sigma_A = M_A - B_A^{\text{rel}}$ (Equation (7.3)). At $a = 0$ the density is uniform, the current F/L is harmonic and $\sigma = F^2/T$ (gate SYM.6). With $a = 1$, $F = 1$, $T = 0.5$, $L = 2\pi$: the closed form is stationary to $2 \cdot 10^{-7}$ on 65 536 points with a spectral derivative, $j = 0.0997$, and on the zone $[0, \pi]$, $\Phi_A^{\text{rel}} = B_A^{\text{rel}} = 0.109$, $\sigma_A = 0.916$, $M_A = 1.025$ (gates RING.1, RING.2; Figure 13).

S.B Proofs

This section proves the theorems of the body from the imported results named in Proposition G1 and in the text; everything else is proved here.

S.B.1 Kernel: retention, sufficiency, the score

Proof of the second claim of Theorem K3, another divergence, another rule (C2). Let f be convex, differentiable on $(0, \infty)$, $f(1) = 0$, $D_f(\mu||\pi) = \sum_i \pi_i f(\mu_i/\pi_i)$ on a simplex with π of full support and likelihood L with positive entries. Suppose the least-deformation update equals $\pi_{\mathcal{D}}$ for every prior of full support and every positive likelihood. The minimizer $\pi_{\mathcal{D}}$ is interior, and the stationarity condition of $\min_{\mu} \{\mathbb{E}_{\mu}[-\log L] + D_f(\mu||\pi)\}$ on the simplex reads $f'(\mu_i/\pi_i) = \log L_i + c$ for every i , with a multiplier c that depends on the instance. At $\mu = \pi_{\mathcal{D}}$, with $u_i = L_i/Z$, it reads $g(u_i) = c + \log Z$ for every i , where $g(u) := f'(u) - \log u$: g takes one value on the ratios of each instance. Take two states, $0 < u < 1 < w$, the prior $(p, 1-p)$ with $p = (w-1)/(w-u)$ and the likelihood (u, w) : then $Z = pu + (1-p)w = 1$, the ratios are u and w , and $g(u) = g(w)$. Fixing $w > 1$ gives $g(u) = g(w)$ for every $u \in (0, 1)$, and fixing $u < 1$ gives $g(w) = g(u)$ for every $w > 1$; f' is continuous, f being convex and differentiable, so g is constant on $(0, \infty)$, $f'(u) = \log u + c'$, $f(t) = t \log t + a(t-1)$ and $D_f = D$ on probabilities. For relative entropy the discrepancy $f'(u) - \log u$ is constant; for the squared Hellinger distance, $f'(u) = 1 - u^{-1/2}$, and for χ^2 , $f'(u) = 2(u-1)$, it varies by 1.85 and $1.8 \cdot 10^3$ in variance on $[10^{-2}, 10^2]$, and the least-deformation update differs from Bayes by 0.03 to 0.13 in total variation on random evidences (gates C2.4b, C2.2; the optimizer control C2.1 finds Bayes to 10^{-9} with relative entropy). $\square$

Proof of Theorem K4, sufficiency and the recovery kernel. When $D(P \parallel R) < \infty$, a zero residual term in Equation (3.2) means $\mathbb{E}_{KP} D(P_{X|Y} \parallel R_{X|Y}) = 0$, that is $P_{X|Y} = R_{X|Y}$ KP -almost surely: the conditional law of the fine variable given the observation is the same under P and under R , which is sufficiency of K for the pair. For the declared family, let L be the Markov kernel $y \mapsto P_{X|Y=y}$, the same for every λ by sufficiency; then $LKP_\lambda = \int P_{X|Y=y} (KP_\lambda)(dy) = P_\lambda$, so L recovers each law of the family from its observed law. Conversely, if such an L exists, the observed experiment K and the original one are each a garbling of the other, so they are equivalent in Blackwell's order and K is sufficient (Blackwell, 1953). On a regular stratum the ledger of Equation (4.2) then has zero residual term, and the Fisher metric of the observed laws equals h . $\square$

Proof of Theorem O3 and of the score projection O2. Let p_θ be dominated and differentiable in θ with a locally dominated derivative, and let K be an observation channel free of θ . Work with the joint law $P_\theta(dx)K(x, dy)$: its score is $S_v(X)$, since K carries zero dependence on θ , so the statement for a channel follows from the statement for the deterministic coordinate of the joint law, and it is enough to treat $Y = q(X)$ with p_θ^Y the density of Y . Differentiating $p_\theta^Y(y) = \int_{q^{-1}(y)} p_\theta d\nu_y$ under the integral gives $\partial_v p_\theta^Y(y) = p_\theta^Y(y) \mathbb{E}_\theta[S_v \mid Y = y]$, that is $S_v^Y = \mathbb{E}[S_v \mid Y]$, Fisher's identity for the score of an observation (O2). With $S_v \in L^2(P_\theta)$ and $\mathbb{E}S_v = 0$, the law of total variance gives $h(v, v) = \mathbb{E}(\mathbb{E}[S_v \mid Y])^2 + \mathbb{E} \text{Var}(S_v \mid Y) = h_Y(v, v) + h_Y^{\text{res}}(v, v)$, the expectation over Y of Louis's identity between observed, complete and missing information (Louis, 1982); polarization gives the bilinear form. The residual term is the second derivative at λ of the residual relative information $\lambda' \mapsto \mathbb{E}_{KP_{\lambda'}} D(P_{\lambda'|Y} \parallel P_{\lambda|Y})$, by Equation (3.2) and Equation (4.1). On the gyrator the variance of the conditional score equals the Fisher metric of the marginal in the five directions to 10^{-14} (gate X1.2). $\square$

S.B.2 Slope geometry

Setting. (Λ, h) is a smooth, connected, complete Riemannian manifold of dimension n , with geodesic distance d_h , volume dV_h , gradient ∇_h and Laplace–Beltrami operator Δ_h . The reference is $m = e^{-V} dV_h$ with $V \in C^2(\Lambda)$, finite or σ -finite, under the Bakry–Émery condition

$$\text{CD}(K, \infty) : \quad \text{Ric}_h + \text{Hess}_h V \geq K h \quad \text{for some } K \in \mathbb{R} \quad (\text{S.B.1})$$

(Bakry and Émery, 1985; von Renesse and Sturm, 2005); this is the contract HR0 of Section 4.2. For $\mu = f m$,

$$\mathcal{H}_m(\mu) := \int_\Lambda f \log f dm, \quad \mathcal{I}_{h,m}(\mu) := \int_\Lambda \|\nabla_h \log f\|_h^2 d\mu = 4 \int_\Lambda \|\nabla_h \sqrt{f}\|_h^2 dm,$$

the second when $\sqrt{f} \in W_{\text{loc}}^{1,2}(m)$ with a finite integral and $+\infty$ otherwise, both $+\infty$ when μ charges a set of m -measure zero. Under Equation (S.B.1) the reference has at most Gaussian volume growth (Sturm, 2006), so that $\mathcal{H}_m$ is well defined with values in $(-\infty, +\infty]$ on the space $\mathcal{P}_2(\Lambda)$ of probability measures with finite second moment (Ambrosio et al., 2008, §9.4). The regular domain is $\mathcal{P}_{\text{reg}} := \{\mu \in \mathcal{P}_2(\Lambda) : \mu = f m, \mathcal{H}_m(\mu) < \infty, \mathcal{I}_{h,m}(\mu) < \infty\}$. $W_{2,h}$ is the quadratic Wasserstein distance of (Λ, d_h) ; a curve $\gamma : [0, 1] \rightarrow \mathcal{P}_2(\Lambda)$ is absolutely continuous (AC) when $W_{2,h}(\gamma_s, \gamma_t) \leq \int_s^t g(r) dr$ for some $g \in L^1(0, 1)$, and its metric derivative $|\dot{\gamma}_t| := \lim_{s \rightarrow t} W_{2,h}(\gamma_s, \gamma_t)/|s - t|$ exists for a.e. t and is the least such g (Ambrosio et al., 2008, §1.1). For a functional F on a metric space (X, d) , the descending metric slope at x with $F(x) < \infty$ is $|\partial F|_d(x) := \limsup_{y \rightarrow x} (F(x) - F(y))^+ / d(x, y)$.

Proposition G1 (The imported Wasserstein calculus of entropy · kernel · IMPORTED). *Under Equation (S.B.1): (i) slope: for $\mu \in \mathcal{P}_{\text{reg}}$, $|\partial \mathcal{H}_m|_{W_{2,h}}^2(\mu) = \mathcal{I}_{h,m}(\mu)$; (ii) strong upper gradient: for every AC curve γ in $\mathcal{P}_{\text{reg}}$ with $\int_0^1 \sqrt{\mathcal{I}_{h,m}(\gamma_t)} |\dot{\gamma}_t| dt < \infty$, the map $t \mapsto \mathcal{H}_m(\gamma_t)$ is absolutely continuous and*

$$\left| \frac{d}{dt} \mathcal{H}_m(\gamma_t) \right| \leq \sqrt{\mathcal{I}_{h,m}(\gamma_t)} |\dot{\gamma}_t| \quad \text{for a.e. } t; \quad (\text{S.B.2})$$

(iii) gradient flow: for $\mu \in \mathcal{P}_{\text{reg}}$ the law $(\mu_t)_{t \geq 0}$ of the m -reversible diffusion with generator $\Delta_h - \nabla_h V \cdot \nabla_h$ started from μ is an AC curve in $\mathcal{P}_{\text{reg}}$, it is the gradient flow of $\mathcal{H}_m$ in $(\mathcal{P}_2(\Lambda), W_{2,h})$, $t \mapsto \mathcal{H}_m(\mu_t)$ is nonincreasing, and for a.e. t

$$\frac{d}{dt} \mathcal{H}_m(\mu_t) = -\mathcal{I}_{h,m}(\mu_t) = -\sqrt{\mathcal{I}_{h,m}(\mu_t)} |\dot{\mu}_t|, \quad |\dot{\mu}_t| = \sqrt{\mathcal{I}_{h,m}(\mu_t)}; \quad (\text{S.B.3})$$

(iv) lower semicontinuity: $\mathcal{H}_m$ is lower semicontinuous on $(\mathcal{P}_2(\Lambda), W_{2,h})$ (Jordan et al., 1998; Otto, 2001; Ambrosio et al., 2008; Erbar, 2010).

Sources. (i) is Otto's identity, proved in the metric setting for λ -convex functionals (Otto, 2001; Ambrosio et al., 2008); under Equation (S.B.1) the entropy is K -convex along Wasserstein geodesics (von Renesse and Sturm, 2005), and the manifold case is treated by Erbar (2010). (ii) is the fact that the slope of a λ -convex lower semicontinuous functional is a strong upper gradient (Ambrosio et al., 2008, §§1.2, 2.4). (iii) is the identification of the heat flow with the entropy gradient flow (Jordan et al., 1998; Otto, 2001; Erbar, 2010), with the energy identity that characterizes gradient flows in metric spaces (Ambrosio et al., 2008, §1.3). (iv) is Ambrosio et al. (2008, §9.4) under the Gaussian volume growth. For a weighted reference, $m = e^{-V} dV_h$, the heat flow of (iii) is the gradient flow of the entropy on $\text{RCD}(K, \infty)$ spaces (Ambrosio et al., 2014). These four facts are the imports of the slope geometry; the logarithmic Sobolev and Talagrand inequalities used below are those of Bakry and Émery (1985) and Otto and Villani (2000).

Length and distance. For an AC curve γ in $\mathcal{P}_{\text{reg}}$, $L_I(\gamma) := \ell_\star \int_0^1 \sqrt{\mathcal{I}_{h,m}(\gamma_t)} |\dot{\gamma}_t| dt \in [0, \infty]$, and $d_I(\mu, \nu) := \inf\{L_I(\gamma) : \gamma \text{ AC in } \mathcal{P}_{\text{reg}}, \gamma_0 = \mu, \gamma_1 = \nu\}$, with $\inf \emptyset = +\infty$ (Definition G2). The integrand is measurable because $\mathcal{I}_{h,m}$ is lower semicontinuous; L_I is invariant under reparametrization, so d_I is symmetric and satisfies the triangle inequality by concatenation. By Proposition G1 (i) the conformal factor is the metric slope of the entropy itself; it vanishes exactly at $\mu = m$ when m is a probability measure.

Proof of Theorem G3 (i), the floor and the horizon. If $L_I(\gamma) = \infty$ the inequality holds. Otherwise Proposition G1 (ii) applies: $t \mapsto \mathcal{H}_m(\gamma_t)$ is absolutely continuous and, integrating Equation (S.B.2),

$$L_I(\gamma) = \ell_\star \int_0^1 \sqrt{\mathcal{I}_{h,m}(\gamma_t)} |\dot{\gamma}_t| dt \geq \ell_\star \int_0^1 \left| \frac{d}{dt} \mathcal{H}_m(\gamma_t) \right| dt \geq \ell_\star |\mathcal{H}_m(\gamma_1) - \mathcal{H}_m(\gamma_0)|.$$

Taking the infimum over curves gives $d_I(\mu, \nu) \geq \ell_\star |\mathcal{H}_m(\nu) - \mathcal{H}_m(\mu)|$: $\mathcal{H}_m$ is $(1/\ell_\star)$ -Lipschitz for d_I . For the horizon, let (μ_k) be a sequence in $\mathcal{P}_{\text{reg}}$ with $\mathcal{H}_m(\mu_k) \rightarrow +\infty$; then $d_I(\mu_0, \mu_k) \rightarrow +\infty$ by the floor. When $\mu_k \rightarrow \delta_\lambda$ in $W_{2,h}$ and $m(\{\lambda\}) = 0$, the Dirac mass charges a set of m -measure zero, so $\mathcal{H}_m(\delta_\lambda) = +\infty$, and Proposition G1 (iv) gives $\liminf_k \mathcal{H}_m(\mu_k) \geq \mathcal{H}_m(\delta_\lambda) = +\infty$, while $W_{2,h}(\mu_0, \mu_k) \rightarrow W_{2,h}(\mu_0, \delta_\lambda) = (\int d_h(x, \lambda)^2 d\mu_0)^{1/2} < \infty$. With an atomic reference, $D(\delta_\lambda \| m) = -\log m(\{\lambda\})$ is finite and certainty is at finite distance: the horizon is a statement of the continuous sector. $\square$

Conformal reliefs. For a lower semicontinuous relief $a : \mathcal{P}_2(\Lambda) \rightarrow [0, \infty]$, write $\mathcal{O}_a := \{0 < a < \infty\}$ and $L_a(\gamma) := \int_0^1 a(\gamma_t) |\dot{\gamma}_t| dt$, $d_a(\mu, \nu) := \inf\{L_a(\gamma)\}$ over AC curves from μ to ν , so that $L_I = L_a$ and $d_I = d_a$ for $a = \ell_\star \sqrt{\mathcal{I}_{h,m}}$. $\mathcal{H}_m$ has uniform slope $1/\ell_\star$ for a when $|\partial \mathcal{H}_m|_{d_a}(\mu) = 1/\ell_\star$ at every $\mu \in \mathcal{O}_a \cap \{0 < \mathcal{I}_{h,m} < \infty\}$.

Proposition L2 (Conformal contraction $\cdot$ kernel $\cdot$ PROVED). *Let a be continuous at $\mu \in \mathcal{O}_a$ in the $W_{2,h}$ topology, with $\mathcal{O}_a$ open. Then $d_a(\mu, \nu) = a(\mu) W_{2,h}(\mu, \nu) (1 + o(1))$ as $W_{2,h}(\mu, \nu) \rightarrow 0$.*

Proof. Fix $\delta \in (0, a(\mu))$ and $r > 0$ such that $B := \{\nu : W_{2,h}(\mu, \nu) < r\} \subset \mathcal{O}_a$ and $|a - a(\mu)| \leq \delta$ on B , and let $W_{2,h}(\mu, \nu) < r(a(\mu) - \delta)/(a(\mu) + \delta)$. Upper bound: $(\mathcal{P}_2(\Lambda), W_{2,h})$ is a geodesic space

since (Λ, d_h) is complete (Ambrosio et al., 2008, ch. 7); a constant-speed geodesic from μ to ν stays in B , so $d_a(\mu, \nu) \leq (a(\mu) + \delta)W_{2,h}(\mu, \nu)$. Lower bound: an AC curve from μ to ν that stays in B has $L_a \geq (a(\mu) - \delta) \int |\dot{\gamma}| \geq (a(\mu) - \delta)W_{2,h}(\mu, \nu)$; one that leaves B has $W_{2,h}$ -length at least r before leaving, hence $L_a \geq (a(\mu) - \delta)r > (a(\mu) + \delta)W_{2,h}(\mu, \nu)$, which is longer than the geodesic. Thus $(a(\mu) - \delta) \leq d_a(\mu, \nu)/W_{2,h}(\mu, \nu) \leq (a(\mu) + \delta)$ for ν close to μ , and δ is arbitrary. $\square$

Proof of Theorem G3 (ii), the constant slope is unique (G4). (a) *Only the entropy slope has uniform slope.* Let a be continuous and positive on the open set $\mathcal{O}_a$. By Proposition L2, d_a and $W_{2,h}$ are locally equivalent at $\mu \in \mathcal{O}_a \cap \mathcal{P}_{\text{reg}}$, so the lim sup defining $|\partial \mathcal{H}_m|_{d_a}(\mu)$ may be taken along $\nu \rightarrow \mu$ in $W_{2,h}$, and

$$|\partial \mathcal{H}_m|_{d_a}(\mu) = \limsup_{\nu \rightarrow \mu} \frac{(\mathcal{H}_m(\mu) - \mathcal{H}_m(\nu))^+}{a(\mu)W_{2,h}(\mu, \nu)(1 + o(1))} = \frac{|\partial \mathcal{H}_m|_{W_{2,h}}(\mu)}{a(\mu)} = \frac{\sqrt{\mathcal{I}_{h,m}(\mu)}}{a(\mu)} \quad (\text{S.B.4})$$

by Proposition G1 (i). Constancy at $1/\ell_\star$ on $\mathcal{O}_a \cap \{0 < \mathcal{I}_{h,m} < \infty\}$ is then exactly $a = \ell_\star \sqrt{\mathcal{I}_{h,m}}$ there. (b) *The entropy slope has uniform slope*, without any continuity assumption. Upper bound: for every $\nu \in \mathcal{P}_{\text{reg}}$ the floor gives $(\mathcal{H}_m(\mu) - \mathcal{H}_m(\nu))^+ \leq d_I(\mu, \nu)/\ell_\star$, hence $|\partial \mathcal{H}_m|_{d_I}(\mu) \leq 1/\ell_\star$. Lower bound: let (μ_t) be the gradient flow of Proposition G1 (iii) from μ with $\mathcal{I}_{h,m}(\mu) > 0$. By Equation (S.B.3), $d_I(\mu, \mu_t) \leq L_I(\mu|_{[0,t]}) = \ell_\star \int_0^t \mathcal{I}_{h,m}(\mu_r) dr = \ell_\star(\mathcal{H}_m(\mu) - \mathcal{H}_m(\mu_t))$, which tends to 0 as $t \rightarrow 0$ since $\mathcal{H}_m \circ \mu$ is continuous. Moreover $\mathcal{H}_m(\mu_t) < \mathcal{H}_m(\mu)$ for every $t > 0$: otherwise $\mathcal{I}_{h,m}(\mu_r) = 0$ for a.e. r in some $(0, t_0)$, so $\log f_r$ is constant on the connected stratum and $\mu_r = c m$; such a probability measure exists exactly when m is finite, and it is then the normalized reference $\bar{m}$, with $\mathcal{I}_{h,m}(\bar{m}) = 0$. For finite m this gives $\mu_r = \bar{m}$ for a.e. $r \in (0, t_0)$, hence for all $r \in (0, t_0]$ by continuity of the curve, and $\mu = \lim_{r \rightarrow 0} \mu_r = \bar{m}$; for infinite m the alternative is empty. Either way it contradicts $\mathcal{I}_{h,m}(\mu) > 0$. So $\mu_t \neq \mu$, $\mu_t \rightarrow \mu$ in d_I , and $(\mathcal{H}_m(\mu) - \mathcal{H}_m(\mu_t))/d_I(\mu, \mu_t) \geq 1/\ell_\star$, whence $|\partial \mathcal{H}_m|_{d_I}(\mu) \geq 1/\ell_\star$. Therefore the relief $\ell_\star \sqrt{\mathcal{I}_{h,m}}$ gives relative entropy the constant metric slope $1/\ell_\star$ at every regular belief with $0 < \mathcal{I}_{h,m} < \infty$, by (b), and a relief continuous at such a belief gives it the slope $1/\ell_\star$ there exactly when it equals $\ell_\star \sqrt{\mathcal{I}_{h,m}}$ there, by (a): the weight is pointwise rigid. The statement is pointwise because $\sqrt{\mathcal{I}_{h,m}}$ is only lower semicontinuous on $(\mathcal{P}_2(\Lambda), W_{2,h})$, and small oscillations of a density make it arbitrarily large; on a smooth finite-dimensional family of regular beliefs, such as the leaves below, where $\mathcal{I}_{h,m}$ is continuous for the topology of the family, (a) applied to the length structure of the family makes the weight unique. $\square$

The radial distance. Let the reference be a probability measure π , so that $\mathcal{H}_\pi = D(\cdot \| \pi)$. A curve *reaches π in relative entropy* when it is defined on $[0, 1)$, AC on every $[0, t]$, and $D(\gamma_t \| \pi) \rightarrow 0$ as $t \rightarrow 1$; its length is $L_I(\gamma) := \lim_{t \rightarrow 1} L_I(\gamma|_{[0,t]})$. For $\mu \in \mathcal{P}_{\text{reg}}$, $\rho_I(\mu) := \inf\{L_I(\gamma) : \gamma \text{ reaches } \pi \text{ in relative entropy, } \gamma_0 = \mu\}$.

Proof of Theorem G3 (iii), the radial identity (G5). Assume the gradient flow (μ_t) of Proposition G1 (iii) started from μ is ergodic in relative entropy, $D(\mu_t \| \pi) \rightarrow 0$. *Lower bound.* Let γ reach π in relative entropy from μ . For $t < 1$ the floor on $[0, t]$ gives $L_I(\gamma|_{[0,t]}) \geq \ell_\star(D(\mu \| \pi) - D(\gamma_t \| \pi))$; letting $t \rightarrow 1$, $L_I(\gamma) \geq \ell_\star D(\mu \| \pi)$. *Upper bound.* Along the flow, Equation (S.B.3) gives for every T

$$\begin{aligned} L_I(\mu|_{[0,T]}) &= \ell_\star \int_0^T \sqrt{\mathcal{I}_{h,\pi}(\mu_t)} |\dot{\mu}_t| dt = -\ell_\star \int_0^T \frac{d}{dt} D(\mu_t \| \pi) dt \\ &= \ell_\star (D(\mu \| \pi) - D(\mu_T \| \pi)) \xrightarrow{T \rightarrow \infty} \ell_\star D(\mu \| \pi). \end{aligned}$$

The reparametrization $t = T/(1 + T)$ turns the flow into a curve on $[0, 1)$ reaching π in relative entropy, with the same length. Hence $\rho_I(\mu) = \ell_\star D(\mu \| \pi)$ and the gradient-flow trace realizes the infimum: it is the radial geodesic. Along every gradient-flow trace, Equation (S.B.3) reads $d\ell_I = |dr|$ for $r(\mu) = \ell_\star D(\mu \| \pi)$: relative entropy is the arc-length coordinate from the reference along the radial geodesics, and its level sets are the calibrated spheres about π . A sufficient

condition. If π satisfies the logarithmic Sobolev inequality $D(\mu\|\pi) \leq \mathcal{I}_{h,\pi}(\mu)/(2K)$ for all $\mu \ll \pi$, which holds under $\text{CD}(K, \infty)$ with $K > 0$ (Bakry and Émery, 1985; Otto and Villani, 2000), then $\frac{d}{dt}D(\mu_t\|\pi) = -\mathcal{I}_{h,\pi}(\mu_t) \leq -2KD(\mu_t\|\pi)$ and Grönwall's lemma gives $D(\mu_t\|\pi) \leq e^{-2Kt}D(\mu\|\pi) \rightarrow 0$; by Talagrand's inequality $W_{2,h}(\mu_t, \pi)^2 \leq \frac{2}{K}D(\mu_t\|\pi) \rightarrow 0$ as well. A metric. Under the logarithmic Sobolev inequality, extend the length structure to $\bar{\mathcal{P}} := \mathcal{P}_{\text{reg}} \cup \{\pi\}$ by admitting the curves that reach π in relative entropy, their reversals and their concatenations; the resulting infimum of lengths $\bar{d}_I$ is symmetric and satisfies the triangle inequality, $\bar{d}_I(\pi, \mu) = \rho_I(\mu)$, $\bar{d}_I(\mu, \nu) \leq \ell_\star(D(\mu\|\pi) + D(\nu\|\pi)) < \infty$ through the centre, and for $\mu \neq \nu$ with $D_\mu := D(\mu\|\pi) > 0$,

$$\bar{d}_I(\mu, \nu) \geq \min \left\{ \frac{1}{2}\ell_\star D_\mu, \ell_\star \sqrt{KD_\mu} W_{2,h}(\mu, \nu) \right\} > 0,$$

because a curve through π has length at least $\ell_\star D_\mu$, while an AC curve in $\mathcal{P}_{\text{reg}}$ from μ to ν of length below $\frac{1}{2}\ell_\star D_\mu$ keeps $D(\gamma_t\|\pi) > \frac{1}{2}D_\mu$ by the floor, hence $\mathcal{I}_{h,\pi}(\gamma_t) > KD_\mu$ along the whole curve by the logarithmic Sobolev inequality, and $L_I(\gamma) \geq \ell_\star \sqrt{KD_\mu} \int_0^1 |\dot{\gamma}_t| dt \geq \ell_\star \sqrt{KD_\mu} W_{2,h}(\mu, \nu)$; so $\bar{d}_I$ separates points and is a metric on $\bar{\mathcal{P}}$. $\square$

Location–scale leaves. Here $\Lambda = \mathbb{R}$ with its Euclidean metric and m is Lebesgue measure ($V = 0$, Equation (S.B.1) with $K = 0$); $\mathcal{H}_m = \text{Ent}(\rho) = \int \rho \log \rho$ and $\mathcal{I}_{h,m} = J(\rho) = \int (\rho'/\rho)^2 \rho$ is the classical Fisher information of a density. Let f be a positive, absolutely continuous probability density with $\int f |\log f| < \infty$, $zf(z) \rightarrow 0$ as $|z| \rightarrow \infty$, centred and standardized, $\int zf = 0$, $\int z^2 f = 1$, $J_0 := \int (f'/f)^2 f < \infty$. Its leaf is $\mathcal{L}_f := \{\rho_{a,s}\}$, $\rho_{a,s}(x) = s^{-1}f((x-a)/s)$, $a \in \mathbb{R}$, $s > 0$, a subset of $\mathcal{P}_{\text{reg}}$.

Proposition L3 (The leaf is a Euclidean half-plane in $W_2 \cdot \text{kernel} \cdot \text{PROVED}$). *For two members of $\mathcal{L}_f$, $W_2(\rho_{a_0,s_0}, \rho_{a_1,s_1})^2 = (a_1 - a_0)^2 + (s_1 - s_0)^2$, the displacement interpolation between them stays in $\mathcal{L}_f$ with a_t, s_t affine in t , and an AC curve $t \mapsto \rho_{a(t),s(t)}$ in $\mathcal{L}_f$ has metric derivative $|\dot{\gamma}_t| = \sqrt{\dot{a}^2 + \dot{s}^2}$.*

Proof. The map $T(x) = a_1 + \frac{s_1}{s_0}(x - a_0)$ is increasing and pushes ρ_{a_0,s_0} forward to ρ_{a_1,s_1} ; in one dimension the increasing push-forward is the quantile coupling, which is optimal for the quadratic cost (Villani, 2009). With $Z \sim f$, $W_2^2 = \mathbb{E}[(a_1 - a_0 + (s_1 - s_0)Z)^2] = (a_1 - a_0)^2 + (s_1 - s_0)^2$ by the standardization. The displacement interpolation $((1-t)\text{id} + tT)_{\#}\rho_{a_0,s_0}$ is ρ_{a_t,s_t} with $a_t = (1-t)a_0 + ta_1$, $s_t = (1-t)s_0 + ts_1$. Hence $(\mathcal{L}_f, W_2)$ is isometric to the Euclidean half-plane and the metric derivative of a curve in the leaf is its Euclidean speed. $\square$

Proof of Theorem G6, hyperbolic leaves. (i) $\partial_x \log \rho_{a,s}(x) = s^{-1}(f'/f)((x-a)/s)$, so $J(\rho_{a,s}) = J_0/s^2$ after the substitution $x = a + sz$. (ii) By Proposition L3 and (i), for an AC curve in the leaf $L_I(\gamma) = \ell_\star \int_0^1 \frac{\sqrt{J_0}}{s(t)} \sqrt{\dot{a}^2 + \dot{s}^2} dt$, the length of the Riemannian metric $G_I|_{\mathcal{L}_f} = \ell_\star^2 J_0 (da^2 + ds^2)/s^2$. Writing it as $e^{2\varphi}(da^2 + ds^2)$ with $\varphi = \frac{1}{2} \log(\ell_\star^2 J_0) - \log s$, its Gaussian curvature is $K = -e^{-2\varphi} \Delta \varphi = -1/(\ell_\star^2 J_0)$. (iii) The metric is the constant multiple $\ell_\star^2 J_0$ of the Poincaré metric $(da^2 + ds^2)/s^2$, whose distance is $\text{arccosh}(1 + |z - w|^2/(2 \text{Im} z \text{Im} w))$ and whose geodesics are the vertical lines and the half-circles orthogonal to the boundary; a constant factor λ on the metric multiplies distances by $\sqrt{\lambda}$ and leaves the geodesics unchanged, so

$$d_{\mathcal{L}_f}((a_0, s_0), (a_1, s_1)) = \ell_\star \sqrt{J_0} \text{arccosh} \left(1 + \frac{(a_1 - a_0)^2 + (s_1 - s_0)^2}{2s_0 s_1} \right).$$

(iv) Since $zf(z) \rightarrow 0$ at infinity, integration by parts gives $\int z f' = -1$; by Cauchy–Schwarz, $1 = (\int z \frac{f'}{f} f)^2 \leq \int z^2 f \cdot \int (\frac{f'}{f})^2 f = J_0$, with equality iff $f'/f = -cz$, i.e. iff f is Gaussian, and $c = 1$ by the normalization (Stam, 1959). Hence $-1/\ell_\star^2 \leq K < 0$ and the Gaussian leaf is the most curved. Along a vertical geodesic from s_0 to s_1 the length is $\ell_\star \sqrt{J_0} |\log(s_1/s_0)|$ while $\text{Ent}(\rho_{a,s}) = \text{Ent}(f) - \log s$ makes the floor equal to $\ell_\star |\log(s_1/s_0)|$: the floor is attained along the vertical geodesics iff $J_0 = 1$, iff f is Gaussian, and then they are global minimizers of d_I , being the

traces of the heat flow of Proposition G1 (iii). On the Gaussian leaf, with the prior $\pi = \mathcal{N}(0, \sigma^2)$ as reference instead, the factor becomes $\mathcal{I}_{h,\pi}(\mathcal{N}(a, s^2)) = a^2/\sigma^4 + (s/\sigma^2 - 1/s)^2$, which vanishes at π : the geometry acquires a centre (Figure 6b). $\square$

Proof of Corollary G7, scale invariance. Let $h' = \alpha h$ with $\alpha > 0$ constant, and either $m' = m$, or $m' = dV_{h'}$ when $m = dV_h$. Then $\nabla_{h'} = \alpha^{-1}\nabla_h$ and $\|\cdot\|_{h'}^2 = \alpha\|\cdot\|_h^2$ give $\|\nabla_{h'} \log f\|_{h'}^2 = \alpha^{-1}\|\nabla_h \log f\|_h^2$, so $\mathcal{I}_{h',m'} = \alpha^{-1}\mathcal{I}_{h,m}$; in the second case $dV_{h'} = \alpha^{n/2}dV_h$, so $f' = \alpha^{-n/2}f$, $\nabla \log f' = \nabla \log f$ and $\mathcal{H}_{m'} = \mathcal{H}_m - \frac{n}{2} \log \alpha$. Distances, hence Wasserstein distances and metric derivatives, scale by $\sqrt{\alpha}$. Then $\sqrt{\mathcal{I}_{h',m'}}|\dot{\gamma}'| = \sqrt{\mathcal{I}_{h,m}}|\dot{\gamma}|$: L_I , d_I , G_I and the differences of $\mathcal{H}$ are invariant, and Equation (S.B.1) holds for (h', m') with $K' = K/\alpha$. When (Λ, h) is the Fisher manifold of an experiment, n independent repetitions turn h into nh . $\square$

The diffusive reference. Let $\varepsilon > 0$ and let the reference diffusion be the m -reversible process $dX_t = -\frac{\varepsilon}{2}\nabla_h V(X_t) dt + \sqrt{\varepsilon} dB_t^h$, with generator $\frac{\varepsilon}{2}(\Delta_h - \nabla_h V \cdot \nabla_h)$, which is non-explosive under Equation (S.B.1) (Bakry and Émery, 1985; Hsu, 2002); its law on $C([0, 1]; \Lambda)$ started from $\mu_0 \in \mathcal{P}_{\text{reg}}$ is R . A controlled diffusion is $dX_t = (-\frac{\varepsilon}{2}\nabla_h V + u_t)(X_t) dt + \sqrt{\varepsilon} dB_t^h$ with the same initial law, u a smooth bounded time-dependent vector field; P is its law and $\mu_t = f_t m$ its marginals, with smooth positive densities. Nelson's current velocity is $v_t := u_t - \frac{\varepsilon}{2}\nabla_h \log f_t$.

Proof of Corollary G8, the diffusive unit. (i) Girsanov's theorem on a manifold (Elworthy, 1982; Hsu, 2002; Léonard, 2012) gives $D(P\|R) = \frac{1}{2\varepsilon} \int_0^1 \int \|u_t\|_h^2 d\mu_t dt$. The Fokker-Planck equation of the controlled process, for densities with respect to dV_h , is $\partial_t \mu = \frac{\varepsilon}{2}[\Delta_h \mu + \nabla_h \cdot (\mu \nabla_h V)] - \nabla_h \cdot (\mu u)$, and $\Delta_h \mu + \nabla_h \cdot (\mu \nabla_h V) = \nabla_h \cdot (\mu \nabla_h \log f)$ since $f = \mu e^V$: this is the continuity equation $\partial_t \mu_t + \nabla_h \cdot (\mu_t v_t) = 0$ with the current velocity. Expanding $\|u\|_h^2 = \|v\|_h^2 + \varepsilon \langle v, \nabla_h \log f \rangle_h + \frac{\varepsilon^2}{4} \|\nabla_h \log f\|_h^2$ and integrating against μ_t , with $\frac{d}{dt} \int f \log f dm = \int \langle v, \nabla_h \log f \rangle_h d\mu$ for smooth positive densities,

$$\varepsilon D(P\|R) = \int_0^1 \left[\frac{1}{2} \int \|\dot{\mu}_t\|_h^2 d\mu_t + \frac{\varepsilon^2}{8} \mathcal{I}_{h,m}(\mu_t) \right] dt + \frac{\varepsilon}{2} [\mathcal{H}_m(\mu_1) - \mathcal{H}_m(\mu_0)]. \quad (\text{S.B.5})$$

(ii) For an AC curve solving the continuity equation with a field v_t , $|\dot{\mu}_t| \leq \|v_t\|_{L^2(\mu_t)}$ for a.e. t , with equality iff v_t is the tangent field, the one of minimal norm (Benamou and Brenier, 2000; Ambrosio et al., 2008); and the elementary $\frac{1}{2}a^2 + b \geq \sqrt{2b}a$ for $a, b \geq 0$, with equality iff $a = \sqrt{2b}$, applied to $a = |\dot{\mu}_t|$ and $b = \frac{\varepsilon^2}{8} \mathcal{I}_{h,m}(\mu_t)$, gives with $\ell_\star = \varepsilon/2$

$$\varepsilon D(P\|R) - \frac{\varepsilon}{2} [\mathcal{H}_m(\mu_1) - \mathcal{H}_m(\mu_0)] \geq \int_0^1 \left[\frac{1}{2} |\dot{\mu}_t|^2 + \frac{\varepsilon^2}{8} \mathcal{I}_{h,m}(\mu_t) \right] dt \geq L_I(\mu), \quad (\text{S.B.6})$$

with equality throughout iff v_t is the tangent field and $\frac{1}{2}|\dot{\mu}_t|^2 = \frac{\varepsilon^2}{8} \mathcal{I}_{h,m}(\mu_t)$ for a.e. t , zero Jacobi energy. (iii) For $u = 0$, $P = R$ and $v = -\frac{\varepsilon}{2}\nabla_h \log f$ is the tangent field of the reference flow, whose speed is $\frac{\varepsilon}{2}\sqrt{\mathcal{I}_{h,m}(\mu_t)}$ by Proposition G1 (iii) after the time change $t \mapsto \varepsilon t/2$: both equalities hold. When m is a probability measure and $\mu_0 = m$, Equation (S.B.6) and the floor give $\varepsilon D(P\|R) \geq \varepsilon D(\mu_1\|m)$, the data-processing inequality for the terminal marginal. The functional $\int [\frac{1}{2}|\dot{\mu}|^2 + c \mathcal{I}_{h,m}]$ with $c = \varepsilon^2/8$ is the action of the Schrödinger problem (Léonard, 2014); its Maupertuis length at energy e is $\int \sqrt{2(e + c \mathcal{I}_{h,m})} |\dot{\mu}|$ (Arnold, 1989), and the slope geometry is its member at zero energy, $\ell_\star^2 = 2c$: a diffusive reference prices every revision above its calibrated length and fixes $\ell_\star = \varepsilon/2$. $\square$

Proof of Corollary G9, Shannon as posterior geometry. Let $\Lambda \sim \pi$ with π satisfying the logarithmic Sobolev inequality, Y an observation with a likelihood and π_Y its posterior, regular for a.e. y . (i) *Lindley.* $I(\Lambda; Y) = \iint \log \frac{dP_{\Lambda Y}}{d(\pi \otimes P_Y)} dP_{\Lambda Y} = \int D(\pi_y\|\pi) P_Y(dy)$ by disintegration, and Theorem G5 for each π_y gives $\mathbb{E}_Y \rho_I(\pi_Y) = \ell_\star I(\Lambda; Y)$. (ii) *Post-processing.* For a channel $Y \rightarrow Z$, $\pi_Z = \mathbb{E}[\pi_Y | Z]$ and the convexity of $D(\cdot\|\pi)$ give $D(\pi_Z\|\pi) \leq \mathbb{E}[D(\pi_Y\|\pi) | Z]$; taking expectations, $\mathbb{E}_Z \rho_I(\pi_Z) \leq \mathbb{E}_Y \rho_I(\pi_Y)$. (iii) *Chain rule.* $I(\Lambda; Y_1, Y_2) = I(\Lambda; Y_1) + I(\Lambda; Y_2 | Y_1)$

with $I(\Lambda; Y_2 | Y_1) = \mathbb{E}_{Y_1} \mathbb{E}_{Y_2|Y_1} D(\pi_{Y_1 Y_2} \| \pi_{Y_1})$: each conditioning re-centres the geometry at the intermediate posterior. (iv) *Rate–distortion*. Let $X \sim \pi$ be a source on Λ , a test channel a Markov kernel producing a reconstruction $\hat{X}$ of law ν , μ_a the law of X given $\hat{X} = a$ (the cells), and $c(x, \hat{x}) = d_h(x, \hat{x})^2$. The tower property gives $\pi = \int \mu_a \nu(da)$; the disintegration of the mutual information gives $I(X; \hat{X}) = \int D(\mu_a \| \pi) \nu(da)$; and $W_{2,h}^2(\mu_a, \delta_a) = \int d_h(x, a)^2 \mu_a(dx)$, the only coupling with a Dirac mass being the product, gives $\mathbb{E} c(X, \hat{X}) = \int W_{2,h}^2(\mu_a, \delta_a) \nu(da)$. With regular cells, Theorem G5 applied to each cell turns Shannon’s $R(D) = \inf\{I(X; \hat{X}) : \mathbb{E} c \leq D\}$ (Shannon, 1959) into

$$\ell_\star R(D) = \inf \left\{ \int \rho_I(\mu_a) \nu(da) : \int \mu_a \nu(da) = \pi, \int W_{2,h}^2(\mu_a, \delta_a) \nu(da) \leq D \right\} :$$

rate is the mean calibrated radius of the cells from the source, distortion the mean transport radius of the cells to their code points. (v) *Zero distortion*. If $\mathbb{E} c = 0$ then $\mu_a = \delta_a$ for ν -a.e. a , so $D(\mu_a \| \pi) = +\infty$ and $I(X; \hat{X}) = +\infty$ for every such channel: $R(0) = +\infty$, and the cells sit on the horizon of Theorem G3. *The Gaussian source*. For $X \sim \mathcal{N}(0, \sigma^2)$ and $0 < D \leq \sigma^2$, the channel with $\hat{X} \sim \mathcal{N}(0, \sigma^2 - D)$ and $X | \hat{X} = a \sim \mathcal{N}(a, D)$ has barycentre $\mathcal{N}(0, \sigma^2)$, cells with $W_2^2(\mathcal{N}(a, D), \delta_a) = D$, and $\mathbb{E}_a \rho_I(\mathcal{N}(a, D)) = \ell_\star \mathbb{E}_a \frac{1}{2} [(a^2 + D)/\sigma^2 - 1 + \log(\sigma^2/D)] = \ell_\star \frac{1}{2} \log(\sigma^2/D) = \ell_\star R(D)$, the Gaussian rate–distortion function (gate S1.1); Shannon’s theorem provides the optimality of this channel, the geometry provides the reading. $\square$

S.B.3 Recovery, stability, closure

Derivation of Realization E1, the balance of a zone and its two ledgers. With $\mu = f m$ and $j = f v$, $\partial_t(f \log f) = (1 + \log f) \partial_t f = -(1 + \log f) \operatorname{div}_m(f v)$; the product rule $\operatorname{div}_m((1 + \log f) f v) = (1 + \log f) \operatorname{div}_m(f v) + f \langle v, \nabla \log f \rangle$ and the divergence theorem for div_m give Equation (7.2) for every declared m . For a diffusion of drift b and diffusion matrix $\mathbf{D}$ with density p relative to the volume, the Fokker–Planck velocity is $v = b - \mathbf{D} \nabla \log p$, so $\langle v, \mathbf{D}^{-1} v \rangle = \langle v, \mathbf{D}^{-1} b \rangle - \langle v, \nabla \log p \rangle$ and, with $\nabla \log p = \nabla \log f + \nabla \log m$, integration over the zone gives Equation (7.3); the reference term vanishes for the volume reference. On the gyrator $\langle v, \nabla \log p \rangle = -x^\top \Gamma^\top C^{-1} x = 0$ pointwise because $\Gamma^\top C^{-1}$ is antisymmetric (gates X4.1b, X4.5); on the driven ring with the Gibbs reference the bulk term changes and the identity holds with its reference term (gate RING.4). $\square$

Proof of Theorem R5, exhaustive resolution recovery. Relative information. Restriction to $\mathcal{F}_N \subset \mathcal{F}_{N+1}$ is a channel, so $D_N \leq D_{N+1} \leq D(P \| R)$ by Equation (3.2). If $P \ll R$, put $Z = dP/dR$ and $Z_N = \mathbb{E}_R[Z | \mathcal{F}_N] = dP|_{\mathcal{F}_N}/dR|_{\mathcal{F}_N}$; Lévy’s upward theorem gives $Z_N \rightarrow Z$ R -almost surely and in $L^1(R)$ (Williams, 1991), and Fatou’s lemma for $\varphi(Z_N) \geq 0$, $\varphi(z) = z \log z - z + 1$, gives $\liminf_N D_N \geq D(P \| R)$; hence $D_N \uparrow D(P \| R)$, and the residual term $D(P \| R) - D_N$ decreases to zero. If P charges an R -null set A , choose $A_N \in \mathcal{F}_N$ with $(P + R)(A \Delta A_N) \rightarrow 0$; the binary data-processing bound $D_N \geq P(A_N) \log \frac{P(A_N)}{R(A_N)} + (1 - P(A_N)) \log \frac{1 - P(A_N)}{1 - R(A_N)}$ tends to $+\infty = D(P \| R)$. This is relative entropy as a supremum over finite partitions (Pinsker, 1964; Barron, 1985). *Fisher*. For a family differentiable in quadratic mean (van der Vaart, 1998, §7), $S_{v,N} = \mathbb{E}[S_v | \mathcal{F}_N]$ is a martingale bounded in L^2 , so $S_{v,N} \rightarrow \mathbb{E}[S_v | \mathcal{F}_\infty] = S_v$ in L^2 ; orthogonality of the increments gives $h_M(v, v) - h_N(v, v) = \mathbb{E}(S_{v,M} - S_{v,N})^2 \geq 0$ and $h_N(v, v) = \|S_{v,N}\|_2^2 \uparrow \|S_v\|_2^2 = h(v, v)$; polarize. *Bayesian information*. The σ -algebras $\mathcal{A} \otimes \mathcal{B}_N$ increase to $\mathcal{A} \otimes \mathcal{B}$ up to null sets; apply the first part to $P = P_{\Lambda Y}$ and $R = \pi \otimes P_Y$. *Radius*. Theorem G3 (iii) and B1 give $\mathbb{E} \rho_{I,N} = \ell_\star I(\Lambda; Y_N)$, and the previous step concludes. For a garbling tower which is a filtration in law only, $K_N P \rightarrow P$, the monotonicity is Blackwell’s and the limit follows from the lower semicontinuity of relative entropy under weak convergence (Ambrosio et al., 2008, Lemma 9.4.3); on the Gaussian sensors of the gyrator the limit is the closed form (gate X2.4). $\square$

Proof of Theorem G10, stability under almost-sufficient projection. (a) For a piecewise C^1 curve c on the stratum, $\sqrt{h_K}(\dot{c}, \dot{c}) \in [\delta, 1] \sqrt{h}(\dot{c}, \dot{c})$ pointwise; integrating and taking the infimum over

curves joining two points gives $\delta d_h \leq d_{h_K} \leq d_h$, and completeness and finite second moments transfer. The couplings of two measures form one set, so the Wasserstein distances are sandwiched in the same way, and so are the difference quotients that define the metric derivative: the absolutely continuous curves coincide and $\delta|\dot{\gamma}|_h \leq |\dot{\gamma}|_{h_K} \leq |\dot{\gamma}|_h$ almost everywhere. (b) $\mathcal{H}_m$ depends on m and μ alone; with $f = d\mu/dm$ one function, $\|\nabla_h \log f\|_h^2 = h^{-1}(d \log f, d \log f)$, and inversion reverses the Loewner order, $h^{-1} \preceq h_K^{-1} \preceq \delta^{-2}h^{-1}$; integrating against μ gives $\mathcal{I}_{h,m} \leq \mathcal{I}_{h_K,m} \leq \delta^{-2}\mathcal{I}_{h,m}$, and the regular domains coincide. (c) Multiply the two sandwiches inside $\ell_\star \int \sqrt{\mathcal{I}} |\dot{\gamma}| dt$ and take the infimum over the common class of curves. (d) Under HR0 for (Λ, h, m) , (c) and the floor of Theorem G3 give $L_1^{h_K}(\gamma) \geq \delta \ell_\star |\mathcal{H}_m(\gamma_1) - \mathcal{H}_m(\gamma_0)|$. (e) Under HR0 for (Λ, h_K, m) the proofs above apply verbatim to (Λ, h_K, m) , and the radial identity has a right-hand side free of the metric. The identification $\delta^2 = 1 - \varphi$ is Theorem 1.4 of Yamaguchi and Nozawa (2024): a statistic is δ -almost sufficient exactly when $h^{\text{res}}(v, v) \leq (1 - \delta^2)h(v, v)$ for every direction, and the largest such ratio is the fraction of missing information of Meng and Rubin (1991). A calibration that changes with the metric, such as the Riemannian volume, is outside the statement. $\square$

Proof of Theorem E6, the predictive closure ledger. Write $X' = X_{t+\Delta}$, $x = x_t$, $X = q(x)$; everything is finite, bounded by $\log |\mathcal{A}|$, and regular conditional laws exist. Since X is $\sigma(x)$ -measurable, $I(X'; x, X) = I(X'; x)$, and the chain rule $I(X'; x, X) = I(X'; X) + I(X'; x | X)$ gives (i) (Cover and Thomas, 2006, Thm 2.5.2); the conditional mutual information equals $\mathbb{E}_x D(P_{X'|x} \| P_{X'|X})$ because $P_{X'|x,X} = P_{X'|x}$ and $P_{X'|X} = \mathbb{E}[P_{X'|x} | X]$, and it vanishes exactly when $P_{X'|x} = P_{X'|X}$ almost surely, which is (ii). Nonnegativity of I_{ret} and I_{res} and $I(X'; x) \leq H(X') = H(X)$, by stationarity, give (iii). Pinsker's inequality for each x and Jensen's inequality for the concave square root give (iv). For (v), $p_\Delta = \mathbb{P}(X' \neq X) \rightarrow 0$ by path continuity, Fano's inequality gives $H(X' | X) \leq h_2(p_\Delta) + p_\Delta \log(|\mathcal{A}| - 1) \rightarrow 0$, hence $I_{\text{ret}} \rightarrow H(X)$, and $I_{\text{ret}} \leq I_{\text{pred}} \leq H(X)$ squeezes I_{pred} and I_{res} . On the empirical measure of a trajectory binned on a grid whose cells refine the symbols, (i) holds exactly, so the numerical content of the gates is in the ratios, the floors by circular shifts and the tightness of (iv) (gates X8.5–X8.8; estimator bias after Paninski (2003)). $\square$

S.C Statements outside the body

Each statement below cites its gates; the theorems of the body are complete without them.

Proposition K6 (The tilt $\cdot$ kernel $\cdot$ IMPORTED). *Every law equivalent to the reference reads $dP = e^V dR$ (Williams, 1991); the potential V is the logarithm of the Radon–Nikodym derivative, and $-\log(dm/d\nu)$ is the reference potential associated with m ; it becomes $\beta E + \text{const}$ in a thermal canonical realization alone (H2).*

Proposition C2 (Least deformation selects Bayes, the controls $\cdot$ kernel $\cdot$ PROVED). *The Gibbs identity of Theorem K3 holds to 10^{-16} on random beliefs, priors and likelihoods on four states (gate C2.3); the stationarity discrepancy $f'(u) - \log u$ is constant for relative entropy and varies for the squared Hellinger distance and for χ^2 (gate C2.4b); the least-deformation update with those two divergences differs from Bayes by 0.03 to 0.13 in total variation, while an optimizer with relative entropy finds Bayes to 10^{-9} (gates C2.2, C2.1).*

Proposition O2 (The score projection $\cdot$ kernel $\cdot$ PROVED). *For an observation $Y = K(X)$ the observable score is the conditional expectation of the fine score, $S_v^Y = \mathbb{E}[S_v | Y]$, and $h_Y(v, w) = \mathbb{E}[S_v^Y S_w^Y]$ (proof in Appendix S.B.1). On the gyrator the variance of the conditional score equals the Fisher metric of the marginal in the five directions to 10^{-14} (gate X1.2).*

Theorem G4 (The constant slope is unique $\cdot$ kernel $\cdot$ PROVED). *The weight $\ell_\star \sqrt{\mathcal{I}_{h,m}}$ gives relative entropy the constant slope $1/\ell_\star$ at every regular belief with $0 < \mathcal{I}_{h,m} < \infty$, and a relief continuous at such a belief gives it the slope $1/\ell_\star$ there exactly when it equals $\ell_\star \sqrt{\mathcal{I}_{h,m}}$ there (Theorem G3,*

proof in Appendix S.B.2). On the relaxation path of the gyrator, the bare transport length is 1.857 and the length with weight $\mathcal{I}_{h,m}$ is 9.15, against a drop of relative information of 3.554 (gate X3.4).

Theorem G5 (The radial identity $\cdot$ kernel $\cdot$ PROVED). *With the prior as reference and an ergodic gradient flow, $\rho_1(\mu) = \ell_\star D(\mu \parallel \pi)$: relative entropy is the radial coordinate and the gradient flow is the radial geodesic (proof in Appendix S.B.2). Physical realization on the gyrator: at equal temperatures $T_1 = T_2 = 1$, from the displaced stationary law $\mathcal{N}(m_0, C)$ with $m_0 = (1.5, -1)$ and C the stationary covariance at $T_1 = 2, T_2 = 0.5$, the relaxation of the ensemble law is a Fokker–Planck flow with isotropic diffusion and a symmetric drift, the gradient flow of $D(\cdot \parallel \pi)$, hence radial, with $\ell_\star = 1$; the calibrated length of the path is 3.55407, the drop of relative information 3.55402, and the entropy production measured on $2 \cdot 10^4$ paths 3.535 ± 0.023 (gates X3.1, X3.2, LIB.8). Figure 6(b) draws the radial geodesics and the calibrated circles on the Gaussian leaf.*

Theorem G6 (Hyperbolic-Leaf Theorem $\cdot$ kernel $\cdot$ PROVED). *With Lebesgue reference, every location-scale family carries the metric $\ell_\star^2 J_0(da^2 + ds^2)/s^2$, a Poincaré half-plane of curvature $-1/(\ell_\star^2 J_0)$ set by the Fisher information of the standardized profile, the Gaussian being extremal by Stam’s inequality (proof in Appendix S.B.2, from four classical ingredients: the leaf is flat and totally geodesic in W_2 , $J(\rho_{a,s}) = J_0/s^2$, the conformal factor of Equation (4.3), and Stam, 1959). The leaves first appear, with the volume as reference, in Caraffa (2026, Thms 4.18–4.19). This hyperbolicity is distinct from the Fisher–Rao hyperbolicity of Gaussian families: the underlying leaf is flat in W_2 (Fukushi et al., 2026), and the negative curvature is created by the factor J_0/s^2 . On the Gaussian leaf the geodesic and the chord of Figure 6 cost 4.5 and 9.6 nats, the curvature is $-1/(\ell_\star^2 J_0)$ symbolically and $J_0 = 1, 2$ for the Gaussian and Laplace profiles (gate S3.1).*

Theorem R2 (The quadruple ledger and its increments $\cdot$ kernel $\cdot$ PROVED). *Along a tower, $D = D_N + D_N^{\text{res}}$, $h = h_N + h_N^{\text{res}}$, $I = I_N + I_N^{\text{res}}$ and $\mathbb{E}\rho_I = \mathbb{E}\rho_{I,N} + \ell_\star I_N^{\text{res}}$: the global, the infinitesimal, the Bayesian and the geometric ledgers agree, each under its own scope (Table 3); from one level to the next the increments are the conditional ledgers, $D_{n+1} - D_n = \mathbb{E} D(P_{Y_{n+1}|Y_n} \parallel R_{Y_{n+1}|Y_n})$, $h_{n+1} - h_n = \mathbb{E}[(S_{n+1} - S_n)^{\otimes 2}]$, $I_{n+1} - I_n = I(\Lambda; Y_{n+1} \mid Y_n)$, $\mathbb{E}\rho_{I,n+1} - \mathbb{E}\rho_{I,n} = \ell_\star I(\Lambda; Y_{n+1} \mid Y_n)$. On nested Gaussian sensors of the gyrator’s state the four quantities increase strictly, $\mathbb{E}\rho_I = \ell_\star I_N$ to 10^{-16} , and the increments equal the conditional quantities to $3 \cdot 10^{-12}$ in absolute value (gates X2.1, X2.8b).*

Proof. The four splits are Equation (3.2), Equation (4.2), the chain rule of mutual information and, with the prior as reference, $\mathbb{E}\rho_{I,N} = \ell_\star I_N$ by B1 and Theorem G3 (iii). The increments are the same identities applied to the pair (Y_n, Y_{n+1}) of a nested tower: the chain rule of relative entropy for the joint law of (Y_n, Y_{n+1}) , the orthogonality of the martingale increments of the score, and $I(\Lambda; Y_{n+1}) = I(\Lambda; Y_n) + I(\Lambda; Y_{n+1} \mid Y_n)$. $\square$

Realization R3 (Harmonic Markov resolution $\cdot$ realization $\cdot$ MEASURED). *Under HR0 with m a probability, let $L_m = \Delta_h + \nabla_h \log w \cdot \nabla_h$ be the reversible generator and $K_t = e^{tL_m}$. Then $(K_t)_{t \geq 0}$ is a Markov semigroup, $K_{t+s} = K_t K_s$, so $t_2 \geq t_1$ makes K_{t_2} a garbling of K_{t_1} ; with a compact resolvent and $-L_m \phi_k = \lambda_k \phi_k$, $K_t \phi_k = e^{-\lambda_k t} \phi_k$; along $\mu_t = K_t \mu$, $-\frac{d}{dt} D(\mu_t \parallel m) = \mathcal{I}_{h,m}(\mu_t)$ (G1): the harmonic spectrum supplies a canonical channel-valued resolution scale, and relative Fisher information is its instantaneous relative-information loss rate. For $d\mu/dm - 1 \in L^2(m)$ the spectral cutoff is the analysis, $\chi^2(\mu \parallel m) = \sum_k a_k^2$ (Parseval) with a tail after diffusion bounded by $e^{-2\lambda_{N+1} t} \|g\|^2$; it is free of any positivity claim, the reconstruction being Realization R4. On the circle the tail bound holds with ratios below 0.06, and the semigroup is positive, mass-preserving, a semigroup to 10^{-16} , diagonal with damping $e^{-k^2 t}$, with $-dD/dt = \mathcal{I}$ to 10^{-7} (gates S2.1, S2.3).*

Realization R4 (The exponential reconstruction $\cdot$ realization $\cdot$ MEASURED). *When the partition function $Z(\theta)$ of the first N modes is finite, the information projection onto their exponential family reconstructs a probability and obeys the Pythagorean identity of Csiszár (1975), $D(\mu \parallel m) = D(\mu \parallel \tilde{\mu}_N) + D(\tilde{\mu}_N \parallel m)$ (Amari and Nagaoka, 2000, §3.2), here on the harmonic realization (gate S2.2). The spectral projector analyses; the exponential projection reconstructs.*

Realization E2 (The interface and the facet ledger $\cdot$ realization $\cdot$ MEASURED). Quantised on a square grid of step ε , the continuum current obeys Kirchhoff on the facets to 10^{-15} ; the finite-volume chain with four neighbours and Scharfetter–Gummel rates reproduces σ with errors 2.1, 0.6, 0.2, 0.1 % at $\varepsilon/\sigma_{\min} = 0.4, 0.2, 0.1, 0.05$; under the tested eight-neighbour mobility convention, the diagonal links create a persistent excess production, above a factor three at every ε (Figure 8; gates X6.1, X6.2, X6.3b). The ledger is a property of the facet conductances of the stencil; entropic gradient-flow structures of discretizations converge to the continuum while their transport distances may converge slowly (Maas, 2011; Forkert et al., 2022).

Realization E3 (Three graphs, and the ruler that bends $\cdot$ realization $\cdot$ MEASURED). The proximity graph (geometric simplices), the generator graph (interaction hyperedges) and the conditional-independence graph (Markov cliques) are three objects. On the gyrator the first is the same lattice at $k = 0$ and $k = 0.6$, while the precision entry $(C^{-1})_{12}$ and the drift entry A_{12} vanish exactly at $k = 0$ (gate X6.5). The law ruler of the stratum $(\log k_1, \log k_2)$ at fixed k is $\frac{1}{2}I$ at $k = 0$, symbolically, and $(C^{-1})_{12} = -k$ times a positive rational function (gate SYM.7); its Gaussian curvature at $(k_1, k_2) = (1, 0.7)$ decreases from 0 at $k = 0$ to -0.274 at $k = 0.6$ and -0.320 at $k = 0.8$, while the partial correlation and the mutual information of the two coordinates increase from zero (gate X6.6; Figure 9).

Realization E4 (The two axes of information $\cdot$ realization $\cdot$ MEASURED). Relative to a product reference, Equation (6.2) holds exactly, the second term being the total correlation. A sensor on one coordinate moves the first part only, a joint sensor the second, and conditioning on one coordinate of a coupled prior drives the coupling part to zero (gates X7.1, X7.2, X7.3).

Realization E7 (The lifetime $\cdot$ realization $\cdot$ MEASURED). The escape rate of a well follows Arrhenius, $e^{-E_b/T}$ (Kramers, 1940): with buffered basin passages the measured rates at $T = 0.25, 0.2, 0.16$ give a log-slope of -0.968 against $1/T$ for a barrier $E_b = 1$ (gate X8.4); a crude zero-crossing count, inflated by recrossings, gives -0.854 (gate PUIITS.2).

S.D Figures and gates

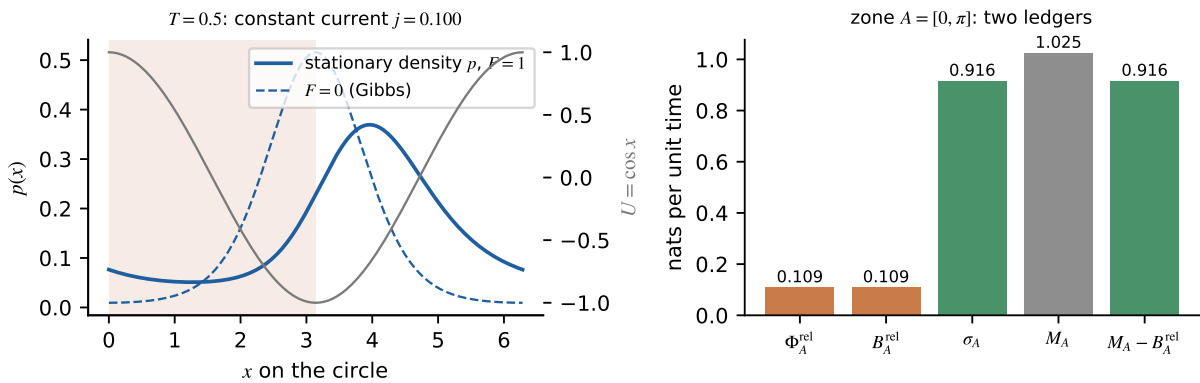


Figure 13: The driven ring. Left: the stationary density (blue) and the potential (grey) for $U = \cos x$, $F = 1$, $T = 0.5$; the current is constant. Right: the ledger of the zone $[0, \pi]$: the interface flux and the bulk term are equal and nonzero, the entropy production equals the medium term minus the bulk term.

The gates follow, grouped by family.

family	what it tests	gates	n
LIB	closed forms of the gyrator	LIB.1, LIB.2, LIB.3, LIB.4, LIB.5, LIB.6, LIB.7, LIB.8	8
PUITS	closed forms of the double well	PUITS.1, PUITS.2	2
SYM	symbolic derivations, gyrator and ring	SYM.1, SYM.2, SYM.3, SYM.4, SYM.5, SYM.6, SYM.7	7
C1	the ledger characterization of relative entropy	C1.1, C1.2	2
C2	the least-deformation characterization of Bayes	C2.1, C2.2, C2.3, C2.4b	4
X1	the identifiable quotient, residual Fisher, almost sufficiency	X1.1, X1.2, X1.3b, X1.4b, X1.5	5
X2	the resolution tower, its limit and its stability	X2.1, X2.2, X2.3, X2.4, X2.5, X2.6, X2.7, X2.8b	8
X3	the calibrated geometry	X3.1, X3.2, X3.3, X3.4	4
X4	the stationary structure, the two ledgers, the control	X4.1b, X4.2, X4.3, X4.4, X4.5	5
X5	the residual arrow under a quotient	X5.1, X5.2, X5.3, X5.4	4
X6	emergent cells, facets and graphs	X6.1, X6.2, X6.3b, X6.4b, X6.5, X6.6	6
X7	the two axes of information	X7.1, X7.2, X7.3	3
X8	predictive closure and symbols	X8.2b, X8.3, X8.4, X8.5, X8.6b, X8.7, X8.8	7
S1	Shannon as posterior geometry	S1.1	1
S2	the harmonic resolution	S2.1, S2.2, S2.3	3
S3	the Gaussian leaf	S3.1	1
RING	the driven ring, its zone and its ports	RING.1, RING.2, RING.3, RING.4	4
every gate green			74

S.E Open problems

Open problem Q1 (The Hodge decomposition of residual irreversibility). Can the residual path-space relative entropy of a coarse-graining be decomposed into contributions of the exact, coexact and harmonic components of the current? The gyrator carries a coexact current, the driven ring a harmonic one.

Open problem Q2 (Beyond uniformly almost-sufficient towers). The radial convergence is acquired by exhaustive recovery (R5) and the non-radial convergence of lengths and distances along uniformly almost-sufficient towers by Corollary G11. What remains: the compactness and convergence of minimizing geodesics, the towers without uniform almost sufficiency, and the finite-state realizations whose coarse metrics leave the smooth stratum, where the evolutionary Γ -convergence of entropic gradient-flow structures (Forkert et al., 2022) is the available tool.

Open problem Q3 (Port references and a common thermodynamic statistic). Under what conditions can a family of port references be represented by a common finite-dimensional thermodynamic statistic, energy being the canonical candidate? The Darmon–Koopman–Pitman theorem is a lead, and the formalization of the port family is open.

Open problem Q4 (Stratified quotients). The operational geometry on quotients whose strata have different dimensions, and its spectral limit. Recent convergence results for graph Dirichlet energies and graph Laplacians on unions of intersecting manifolds of varying dimensions (Bungert and Slepčev, 2026) suggest an analytic backend once the metric, the measure and the transmission conditions across strata are fixed; the operational problem is to derive those conditions from the channel quotient.

Open problem Q5 (Composing phenomena of different natures in one system). Two informational dissipative structures of different natures, coupled through a shared port: do their three ledgers close together as one structure, and does consistency across scales compose along the two towers? This paper reads three benchmarks of different natures with the same ledger, separately; their coupled composition awaits measurement.

S.F Notation

$D(P \parallel R)$	relative information of P from the declared reference R , in nats
$\mathfrak{J} = (\mathcal{M}_{\text{adm}}, \mathcal{S}, \mathcal{E}, R, \pi; m_\Lambda)$	the datum: candidates, semantics, operational access, path reference, prior, calibration
$p_t \in \mathcal{P}(\mathbf{X}), m_{\mathbf{X}}$	ensemble law of a physical state and its reference (K0)
$\bar{\Lambda}_{\mathcal{E}}, h$	identifiable quotient and its Fisher metric (K1, O1)
$\mu \in \mathcal{P}(\bar{\Lambda}_{\mathcal{E}}), \pi_Y, I(\Lambda; Y)$	belief, posterior, mutual information (B1)
$\mathcal{H}_m, \mathcal{I}_{h,m}, W_{2,h}, d\ell_1, \ell_\star$	relative entropy, relative Fisher information, transport distance, calibrated length, gauge (G1, G2)
Σ, σ, Q_i	path irreversibility, entropy production rate, heat delivered to bath i (K5, H2)
$A, \mathbf{D}, C, \Gamma, s$	drift, diffusion matrix, stationary covariance, current matrix and vorticity of the gyrator (H1)
$K_N, D_N, D_N^{\text{res}}, h_N, I_N, \rho_I$	resolution, visible and residual ledgers, calibrated radius (R1, R2)
$I_{\text{pred}}, I_{\text{ret}}, I_{\text{res}}, \varepsilon_{\text{cl}}, \eta_{\text{mem}}$	predictive, retained, residual information of a quotient at time resolution Δ , loss and memory (E6)
$\Phi_A^{\text{rel}}, B_A^{\text{rel}}, \sigma_A, M_A$	interface flux, bulk term, entropy production and medium term of a zone (E1)
$k_B, \dot{W}, \dot{Q}_i$	Boltzmann's constant, set to one in the units of the gyrator; work rate received from the work ports, heat rate delivered to bath i (H2)
